



Aegis AeroAssure®

AUDIT DEFENSE PACK

AS9100D / ISO 9001:2015

Meridian Aerostructures

Wichita, Kansas

Scope: Design-to-print manufacture of aerostructure details and sub-assemblies

Audit Type: Registrar

Generated from data as of: 2026-10-08 21:29 UTC

Prepared by: Aegis AeroAssure (Recovery consultant)

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1. Executive Summary

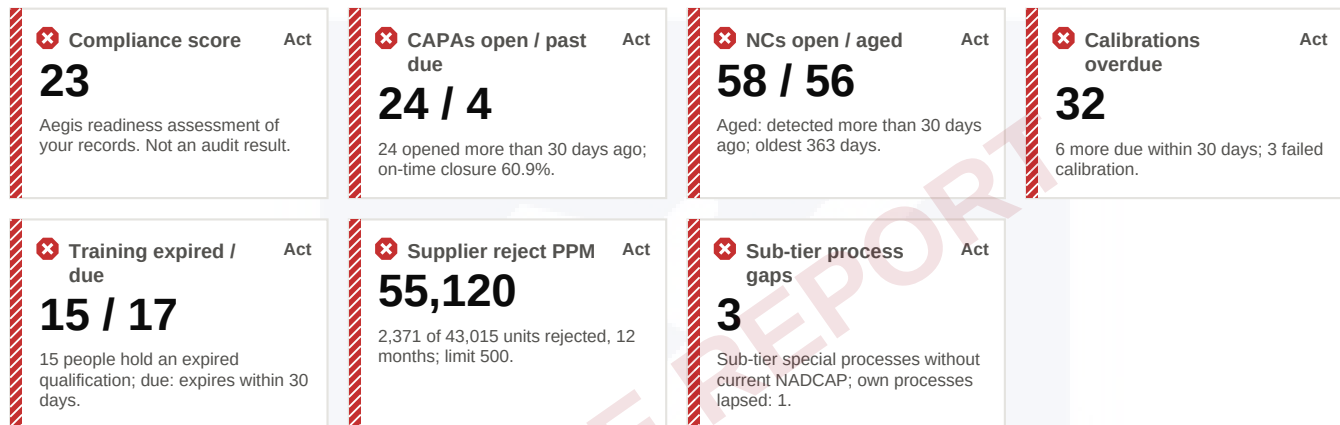
61 issues ranked for action from the records; 10 groups of them share one cause.

The AS9100D compliance score is 23. 144 findings open across audits, nonconformances and self-assessment; 4 CAPAs past their due date; 32 measuring instruments past calibration; 15 people with an expired qualification.

QMS snapshot

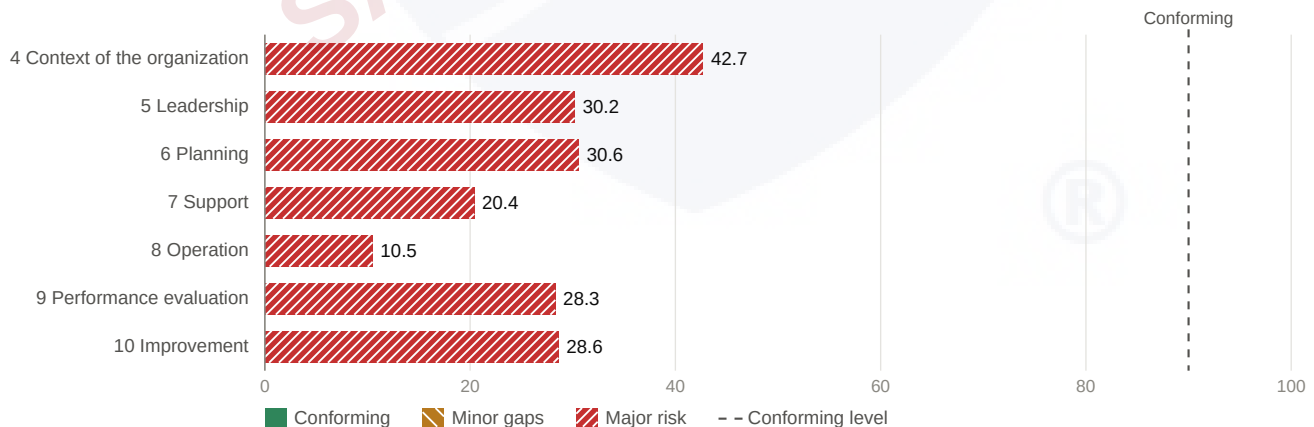
Every number below is a key figure counted from the registers on 2026-10-08; the key figures table that follows says what each measures.

Headline figures



Compliance score: 23 (act, Aegis readiness assessment of your records. Not an audit result.); CAPAs open / past due: 24 / 4 (act, 24 opened more than 30 days ago; on-time closure 60.9%.); NCs open / aged: 58 / 56 (act, Aged: detected more than 30 days ago; oldest 363 days.); Calibrations overdue: 32 (act, 6 more due within 30 days; 3 failed calibration.); Training expired / due: 15 / 17 (act, 15 people hold an expired qualification; due: expires within 30 days.); Supplier reject PPM: 55,120 (act, 2,371 of 43,015 units rejected, 12 months; limit 500.); Sub-tier process gaps: 3 (act, Sub-tier special processes without current NADCAP; own processes lapsed: 1.).

Clause scores



Top-level AS9100D clause scores: Aegis's proprietary assessment of the documents, evidence and open findings recorded against each clause. Not an audit result, and not a certification decision.

Clause scores: 4 42.7, 5 30.2, 6 30.6, 7 20.4, 8 10.5, 9 28.3, 10 28.6. Major risk: 4 (42.7), 5 (30.2), 6 (30.6), 7 (20.4), 8 (10.5), 9 (28.3), 10 (28.6).

Every QMS element at a glance

	Current	⚠ Due in 30 days	✖ Overdue	✖ Missing
Documents	23	3	5	7
Records	115	n/a	n/a	11
Training	35	17	15	5
Calibration	83	6	32	0
CAPA	7	13	4	0
Nonconformances	7	10	41	0
Suppliers	0	0	12	13
Sub-tier and special processes	13	1	2	2
First article (AS9102)	0	4	11	3
Internal audits	8	0	0	0
Risk and PFMEA	4	6	0	15
Management review	2	0	0	0

Count: 0 1 to 115, darker is more

Each QMS element's records by state on the report date. Current: in date, or not yet due. n/a: the state does not apply to that element, or its records were not read.

Documents: 23 current, 3 due in 30 days, 5 overdue, 7 missing; Records: 115 current, 11 missing; Training: 35 current, 17 due in 30 days, 15 overdue, 5 missing; Calibration: 83 current, 6 due in 30 days, 32 overdue, 0 missing; CAPA: 7 current, 13 due in 30 days, 4 overdue, 0 missing; Nonconformances: 7 current, 10 due in 30 days, 41 overdue, 0 missing; Suppliers: 0 current, 0 due in 30 days, 12 overdue, 13 missing; Sub-tier and special processes: 13 current, 1 due in 30 days, 2 overdue, 2 missing; First article (AS9102): 0 current, 4 due in 30 days, 11 overdue, 3 missing; Internal audits: 8 current, 0 due in 30 days, 0 overdue, 0 missing; Risk and PFMEA: 4 current, 6 due in 30 days, 0 overdue, 15 missing; Management review: 2 current, 0 due in 30 days, 0 overdue, 0 missing.

Documents: In force and approved, review date more than 30 days off; review due within 30 days; review date passed; not approved or no review date. Records: Closed records naming where their evidence is kept; those naming none are missing. Training: Qualifications current; expiring within 30 days; expired; assigned and not started. Calibration: Gauges in service due after 30 days; due within 30; past due; no due date. CAPA: Open CAPAs by due date: more than 30 days off; within 30; past due; no due date. Nonconformances: Open nonconformances by due date, as CAPAs. Suppliers: Re-audited within a year; re-audit due within 30 days; past due or under watch (probation, conditional, suspended); never audited. Sub-tier and special processes: Own processes and sub-tier scope: current; approval expiring within 30 days; expired or lapsed; no NADCAP accreditation. First article (AS9102): Signed, or not yet due; unsigned and due within 30 days; unsigned past due; recorded Accepted with no signatory. Internal audits: Internal audits completed or scheduled more than 30 days off; scheduled within 30 days; scheduled date passed, not completed; no date. Risk and PFMEA: Open risks by due date; missing counts high-priority failure modes no control plan names. Management review: Reviews held in the last year; scheduled within 30 days; one missing when none was held in the year.

KPI trend



On-time delivery 54% in 2026-W39 against a target of 90%, off target; First-pass yield 64% in 2026-W39 against a target of 92%, off target; CAPA closure 60.9% in 2026-W39 against a target of 95%, off target; Scrap 8.2% in 2026-W39 against a target of 2%, off target.

Key figures

Measure	Value	What it measures
AS9100D compliance score	23	Aegis readiness score: a proprietary assessment of the documents, evidence and open findings recorded against each AS9100D clause. Not an audit result, and not a certification decision.
Open findings, all sources	144 (12 audit findings, 58 nonconformances, 74 assessment nonconformities)	Open audit findings, open nonconformances and nonconforming self-assessment answers, together.
Open nonconformances	58	Nonconformances not yet closed, at any stage from open to verification.
Oldest open nonconformance (days since detected)	363	Days from the detected date of the oldest open nonconformance to the report date.
Open CAPAs	24	CAPAs not closed or cancelled.
Open CAPAs past their due date	4	Open CAPAs whose due date is before the report date.
Closed CAPAs	46 (37 effectiveness verified, 9 effectiveness not yet verified)	CAPAs closed, with or without effectiveness verified.
CAPA on-time closure (%)	60.9	28 of 46 closed CAPAs with a due date were closed on or before it.
Measuring equipment past calibration due date	32	Equipment whose next calibration date has passed.
Equipment that failed calibration	3	Calibration failures with an impact evaluation raised.
People with an expired qualification	15	Distinct people holding at least one expired qualification.
Training assignment completion (%)	65.3	47 of 72 training assignments completed.
First article reports lacking signatures	18 (15 unsigned, 3 recorded Accepted with no signatory)	AS9102 reports without a reviewer and an approver signature recorded: those not signed, and those the client's log records as Accepted with no signatory.
Part submission warrants not signed	6	PPAP submissions whose warrant is not signed.
Own special processes without a current approval	1	The organisation's own Nadcap-accredited processes whose accreditation is not current, and customer-approved processes whose approval has expired. Sub-tier suppliers' processes are counted separately.
Sub-tier special processes without a current NADCAP accreditation	3	Special processes a supplier's sub-tier performs with an expired NADCAP accreditation or none for that process, from the sub-tier map.
Customer requirements not flowed down	9	Accepted customer requirements with nothing recorded as flowed to the supplier or the internal process they apply to (no mapping, or only a gap).
Customer requirements only partly flowed down	4	Accepted customer requirements recorded as partly flowed: started, not complete. Counted apart from those flowed down and those not flowed down at all.
Open counterfeit-part cases	3	Counterfeit or suspect-part cases not closed.
Suppliers on probation, conditional or disqualified	12	Suppliers whose status is probation, conditional, disqualified or suspended.

Priority actions

Most pressing first, in the order of Aegis's proprietary prioritisation of your findings. The same records always give the same order.

#	Action	Why now	Owner	Clause
1	Supplier Great Plains Titanium (4 findings) - Quarantine the material in counterfeit case CF-MER-003, trace where it went, and notify the customers whose parts share the route - Require Great Plains Titanium to send nondestructive testing to a NADCAP-accredited source instead of Wichita Heat Treating Inc, or show Wichita Heat Treating Inc's current accreditation, before accepting more parts - Flow down FLD-MER-01 (Boeing) to Great Plains Titanium, and record the supplier's acknowledgement - Confirm with Great Plains Titanium that Central NDT Services renews NADCAP NAD-NDT-11042 for nondestructive testing before 2026-10-15	Supplier Great Plains Titanium appears in 4 separate findings across flow-down, special processes, traceability. These sit in different registers and read as unrelated; together they describe one source that is not under control, and addressing them as one supplier action closes all 4.	Not assigned	8.1.4, 8.4.2, 8.4.3
2	Part MER-3101 (5 findings) - Complete the open PPAP elements and sign the part submission warrant for MER-3101 (PPAP-MER-3101) - Qualify the measurement system (Gage R&R) before relying on the capability result: Bore diameter on MER-3101 measured with BG-14 - Qualify the measurement system (Gage R&R) before relying on the capability result: Flange thickness on MER-3101 measured with MIC-7015 - Add the High-priority failure modes on MER-3101 (PFMEA-MER-3101) to the control plan, each with a control method and reaction plan - Complete the reviewer and approver signatures on the first article report for MER-3101 (FAI-MER-101)	Part MER-3101 carries 5 findings across APQP, capability, first article inspection. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 5 independent lapses.	Not assigned	7.1.5.2, 8.1.1, 8.5.1.3
3	Supplier Aero Fastener Brokers (3 findings) - Quarantine the material in counterfeit case CF-MER-001, trace where it went, and notify the customers whose parts share the route - Quarantine the material in counterfeit case CF-MER-002, trace where it went, and notify the customers whose parts share the route - Complete the flow-down of FLD-MER-16 (Boeing) to Aero Fastener Brokers, and record the evidence	Supplier Aero Fastener Brokers appears in 3 separate findings across flow-down, traceability. These sit in different registers and read as unrelated; together they describe one source that is not under control, and addressing them as one supplier action closes all 3.	Not assigned	8.1.4, 8.4.3

#	Action	Why now	Owner	Clause
4	Part MER-3102 (3 findings) • Complete the open PPAP elements and sign the part submission warrant for MER-3102 (PPAP-MER-3102) • Add the High-priority failure modes on MER-3102 (PFMEA-MER-3102) to the control plan, each with a control method and reaction plan • Complete the reviewer and approver signatures on the first article report for MER-3102 (FAI-MER-102)	Part MER-3102 carries 3 findings across APQP, first article inspection. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 3 independent lapses.	Not assigned	8.1.1, 8.5.1.3
5	Part MER-3103 (3 findings) • Complete the open PPAP elements and sign the part submission warrant for MER-3103 (PPAP-MER-3103) • Add the High-priority failure modes on MER-3103 (PFMEA-MER-3103) to the control plan, each with a control method and reaction plan • Complete the reviewer and approver signatures on the first article report for MER-3103 (FAI-MER-103)	Part MER-3103 carries 3 findings across APQP, first article inspection. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 3 independent lapses.	Not assigned	8.1.1, 8.5.1.3
6	Supplier Great Bend Machining (2 findings) • Require Great Bend Machining to send coatings to a NADCAP-accredited source instead of Plains Surface Technologies, or show Plains Surface Technologies's current accreditation, before accepting more parts • Flow down FLD-MER-13 (Spirit) to Great Bend Machining, and record the supplier's acknowledgement	Supplier Great Bend Machining appears in 2 separate findings across flow-down, special processes. These sit in different registers and read as unrelated; together they describe one source that is not under control, and addressing them as one supplier action closes both.	Not assigned	8.4.2, 8.4.3
7	Part MER-3104 (3 findings) • Complete the open PPAP elements and sign the part submission warrant for MER-3104 (PPAP-MER-3104) • Add the High-priority failure modes on MER-3104 (PFMEA-MER-3104) to the control plan, each with a control method and reaction plan • Record who reviewed and approved the accepted first article report for MER-3104 (FAI-MER-104)	Part MER-3104 carries 3 findings across APQP, first article inspection. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 3 independent lapses.	Not assigned	8.1.1, 8.5.1.3
8	Part MER-3106 (3 findings) • Add the High-priority failure modes on MER-3106 (PFMEA-MER-3106) to the control plan, each with a control method and reaction plan • Complete the reviewer and approver signatures on the first article report for MER-3106 (FAI-MER-106) • Complete the open PPAP elements and sign the part submission warrant for MER-3106 (PPAP-MER-3106)	Part MER-3106 carries 3 findings across APQP, first article inspection. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 3 independent lapses.	Not assigned	8.1.1, 8.5.1.3

#	Action	Why now	Owner	Clause
9	Part MER-3105 (3 findings) • Add the High-priority failure modes on MER-3105 (PFMEA-MER-3105) to the control plan, each with a control method and reaction plan • Complete the open PPAP elements and sign the part submission warrant for MER-3105 (PPAP-MER-3105) • Record who reviewed and approved the accepted first article report for MER-3105 (FAI-MER-105)	Part MER-3105 carries 3 findings across APQP, first article inspection. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 3 independent lapses.	Not assigned	8.1.1, 8.5.1.3
10	Part MER-3107 (2 findings) • Qualify the measurement system (Gage R&R) before relying on the capability result: Hardness HRC on MER-3107 measured with HT-2205 • Complete the reviewer and approver signatures on the first article report for MER-3107 (FAI-MER-107)	Part MER-3107 carries 2 findings across capability, first article inspection. A single part accumulating issues in several controls usually means the process was released before it was ready, rather than 2 independent lapses.	Not assigned	7.1.5.2, 8.5.1.3
11	Stop releasing work through NDT — Radiographic Inspection until its approval is current, or move it to an approved source	NDT — Radiographic Inspection is performed without current NADCAP accreditation (PRI, expiry 2026-09-14). A special process is one whose result cannot be verified by inspection afterwards, which is precisely why the approval is the control.	Not assigned	8.5.1.2
12	Require Heartland Aluminum Corp to send chemical processing to a NADCAP-accredited source instead of Flint Chemical Processing, or show Flint Chemical Processing's current accreditation, before accepting more parts	Flint Chemical Processing, a sub-tier of Heartland Aluminum Corp, performs chemical processing on accreditation NAD-CP-30911, which expired 2026-08-31. Parts that pass through it reach this plant through Heartland Aluminum Corp, so the control lapses here as well.	Not assigned	8.4.2
13	Finish the impact evaluation for Wilson Rockwell Hardness Tester (CAL-MER-02), Fluke 1586A Super-DAQ Pyrometer (CAL-MER-03), Bore Gauge 12mm (cell 4) (CAL-MER-121): identify product accepted on its measurements since the last good calibration and disposition it	3 records; for example: Wilson Rockwell Hardness Tester (CAL-MER-02) failed calibration and the impact evaluation is not complete. Every measurement taken with it since its last passing calibration is in doubt. Product accepted on those measurements has not been dispositioned.	Not assigned	7.1.5.2
14	Implement FLD-MER-02 (Boeing) in Receiving Inspection, FLD-MER-03 (Airbus) in Heat Treat, FLD-MER-04 (Airbus) in Inspection, FLD-MER-05 (GE) in Heat Treat + Inspection, FLD-MER-12 (Airbus) in FAI, and record the evidence	5 records; for example: GE requirement FLD-MER-05 — Product acceptance — hardness verification per AMS 6415 — is not recorded as in place in Heat Treat + Inspection: Hardness tester overdue on calibration (CAL-MER-02). A requirement the organisation took on and did not put in place is a requirement nobody is meeting.	Not assigned	7.1.5.1, 8.5.1, 8.5.2, 8.6
15	Flow down FLD-MER-11 (Airbus) to Titan Bar & Billet, FLD-MER-15 (GE) to Emporia Anodizing Co, and record the supplier's acknowledgement	2 records; for example: Airbus requirement FLD-MER-11 — Key characteristic flowdown — is not recorded as flowed to Titan Bar & Billet: Key characteristics not identified on POs to Titan Bar & Billet. A requirement the organisation took on and did not pass on is a requirement nobody is meeting.	Not assigned	8.4.3
16	Re-plan or escalate CAPA-MER-029, CAPA-MER-042, CAPA-MER-052, CAPA-MER-019: past due and still open	4 records; the most overdue: CAPA-MER-029 (Missing inspection stamp — MER-3105) is 287 days past its due date and still open. An overdue corrective action is the finding an auditor uses to question whether the system corrects anything.	T. Nakamura, R. Vasquez, P. Green	10.2

#	Action	Why now	Owner	Clause
17	Complete the reviewer and approver signatures on the first article report for MER-3108 (FAI-MER-108), MER-3110 (FAI-MER-110), MER-3111 (FAI-MER-111), MER-3112 (FAI-MER-112), MER-3113 (FAI-MER-113), MER-3114 (FAI-MER-114), MER-3115 (FAI-MER-115), MER-3116 (FAI-MER-116), MER-3117 (FAI-MER-117), MER-3118 (FAI-MER-118)	10 records; for example: The FAIR FAI-MER-117 for MER-3117 has no completed reviewer and approver signature. Production process verification is not complete until both are recorded, and they cannot be the same person.	Not assigned	8.5.1.3
18	Complete the flow-down of FLD-MER-09 (Airbus) in Heat Treat, FLD-MER-10 (GE) in All special processes, FLD-MER-18 (Airbus) in Calibration, and record the evidence	3 records; for example: Airbus requirement FLD-MER-18 — Gauge calibration traceable to national standards — is recorded as only partly flowed in Calibration. Part of a requirement passed on is the rest of it not met.	Not assigned	8.1
19	Record who reviewed and approved the accepted first article report for MER-3109 (FAI-MER-109)	The FAIR FAI-MER-109 for MER-3109 is recorded as accepted, but no reviewer or approver is recorded against it. An auditor will ask who signed it.	Not assigned	8.5.1.3

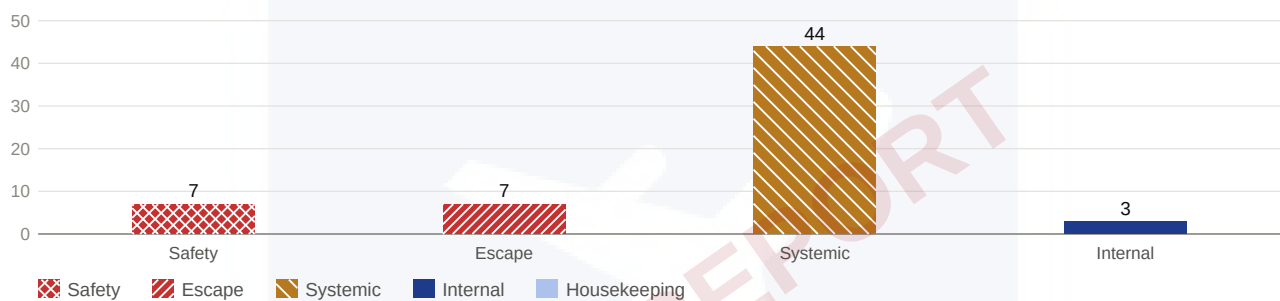
Appendix D turns these into the auditor's own questions: 106 questions across the 15 clauses where your records show a gap, with the evidence to have ready for each.

What these have in common

Shared cause	What closes it	Areas
Supplier Great Plains Titanium (4 findings)	- Quarantine the material in counterfeit case CF-MER-003, trace where it went, and notify the customers whose parts share the route - Require Great Plains Titanium to send nondestructive testing to a NADCAP-accredited source instead of Wichita Heat Treating Inc, or show Wichita Heat Treating Inc's current accreditation, before accepting more parts - Flow down FLD-MER-01 (Boeing) to Great Plains Titanium, and record the supplier's acknowledgement - Confirm with Great Plains Titanium that Central NDT Services renews NADCAP NAD-NDT-11042 for nondestructive testing before 2026-10-15	Flow-down, special processes, traceability
Part MER-3101 (5 findings)	- Complete the open PPAP elements and sign the part submission warrant for MER-3101 (PPAP-MER-3101) - Qualify the measurement system (Gage R&R) before relying on the capability result: Bore diameter on MER-3101 measured with BG-14 - Qualify the measurement system (Gage R&R) before relying on the capability result: Flange thickness on MER-3101 measured with MIC-7015 - Add the High-priority failure modes on MER-3101 (PFMEA-MER-3101) to the control plan, each with a control method and reaction plan - Complete the reviewer and approver signatures on the first article report for MER-3101 (FAI-MER-101)	APQP, capability, first article inspection
Supplier Aero Fastener Brokers (3 findings)	- Quarantine the material in counterfeit case CF-MER-001, trace where it went, and notify the customers whose parts share the route - Quarantine the material in counterfeit case CF-MER-002, trace where it went, and notify the customers whose parts share the route - Complete the flow-down of FLD-MER-16 (Boeing) to Aero Fastener Brokers, and record the evidence	Flow-down, traceability
Part MER-3102 (3 findings)	- Complete the open PPAP elements and sign the part submission warrant for MER-3102 (PPAP-MER-3102) - Add the High-priority failure modes on MER-3102 (PFMEA-MER-3102) to the control plan, each with a control method and reaction plan - Complete the reviewer and approver signatures on the first article report for MER-3102 (FAI-MER-102)	APQP, first article inspection
Part MER-3103 (3 findings)	- Complete the open PPAP elements and sign the part submission warrant for MER-3103 (PPAP-MER-3103) - Add the High-priority failure modes on MER-3103 (PFMEA-MER-3103) to the control plan, each with a control method and reaction plan - Complete the reviewer and approver signatures on the first article report for MER-3103 (FAI-MER-103)	APQP, first article inspection
Supplier Great Bend Machining (2 findings)	- Require Great Bend Machining to send coatings to a NADCAP-accredited source instead of Plains Surface Technologies, or show Plains Surface Technologies's current accreditation, before accepting more parts - Flow down FLD-MER-13 (Spirit) to Great Bend Machining, and record the supplier's acknowledgement	Flow-down, special processes
Part MER-3104 (3 findings)	- Complete the open PPAP elements and sign the part submission warrant for MER-3104 (PPAP-MER-3104) - Add the High-priority failure modes on MER-3104 (PFMEA-MER-3104) to the control plan, each with a control method and reaction plan - Record who reviewed and approved the accepted first article report for MER-3104 (FAI-MER-104)	APQP, first article inspection
Part MER-3106 (3 findings)	- Add the High-priority failure modes on MER-3106 (PFMEA-MER-3106) to the control plan, each with a control method and reaction plan - Complete the reviewer and approver signatures on the first article report for MER-3106 (FAI-MER-106) - Complete the open PPAP elements and sign the part submission warrant for MER-3106 (PPAP-MER-3106)	APQP, first article inspection

Shared cause	What closes it	Areas
Part MER-3105 (3 findings)	• Add the High-priority failure modes on MER-3105 (PFMEA-MER-3105) to the control plan, each with a control method and reaction plan • Complete the open PPAP elements and sign the part submission warrant for MER-3105 (PPAP-MER-3105) • Record who reviewed and approved the accepted first article report for MER-3105 (FAI-MER-105)	APQP, first article inspection
Part MER-3107 (2 findings)	• Qualify the measurement system (Gage R&R) before relying on the capability result: Hardness HRC on MER-3107 measured with HT-2205 • Complete the reviewer and approver signatures on the first article report for MER-3107 (FAI-MER-107)	Capability, first article inspection

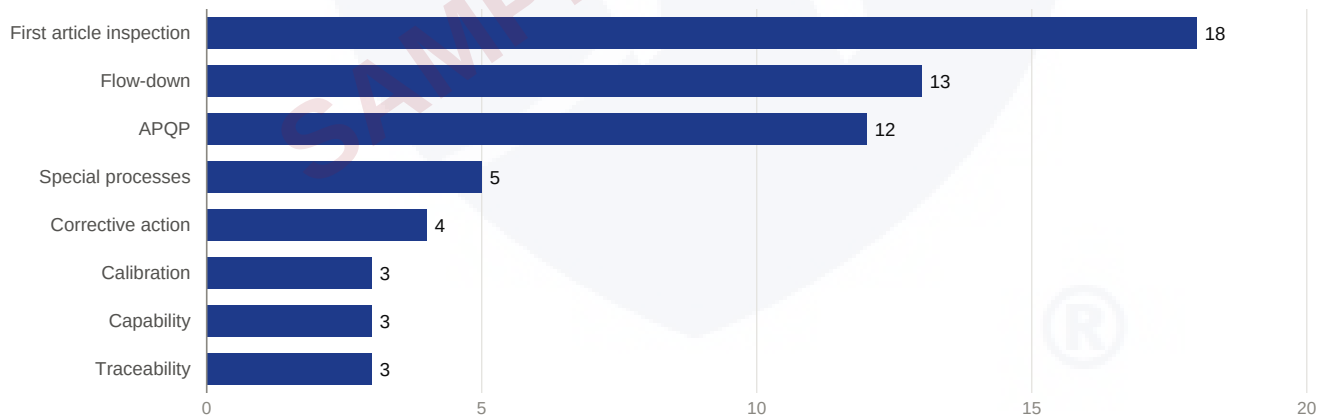
Issues ranked, by severity



Issues ranked, by severity: the ranked findings this section lists, by the worst consequence each could have.

61 ranked issues: 7 safety, 7 escape, 44 systemic, 3 internal.

Issues ranked, by area



Issues ranked, by area of the quality system, most first.

By area: First article inspection 18, Flow-down 13, APQP 12, Special processes 5, Corrective action 4, Calibration 3, Capability 3, Traceability 3.

Every figure and citation is drawn from your records; the priority order is Aegis's assessment.

Where the system is drifting

41 nonconformances past their due date — the worst is 333 days past due (clause 8.7).

#	Open lapse	Why it leads	Severity	Clause
1	Nonconformances past their due date	41 records against clause 8.7; the worst is 333 days past due	High	8.7
2	Corrective actions past their due date	4 records against clause 10.2; the worst is 287 days past due	High	10.2
3	Measuring equipment past its calibration due date	32 records against clause 7.1.5.2; the worst is 170 days past due	Critical	7.1.5.2
4	Assigned training past its due date	12 records against clause 7.2; the worst is 160 days past due	Medium	7.2
5	Documents waiting on approval	7 records against clause 7.5.2; how long each has waited is not recorded	Low	7.5.2



2. Audit Scope & QMS Context

Scope Statement: Design-to-print manufacture of aerostructure details and sub-assemblies

Process landscape

Clause 4: Context of the Organization

- 4.1 Understanding the organization and its context
- 4.2 Understanding the needs and expectations of interested parties
- 4.3 Determining the scope of the QMS
- 4.4 Quality management system and its processes

Clause 6: Planning

- 6.1 Actions to address risks and opportunities
- 6.2 Quality objectives and planning to achieve them
- 6.3 Planning of changes

Clause 8: Operation

- 8.1 Operational planning and control
 - 8.1.1 Operational risk management
 - 8.1.2 Configuration management
 - 8.1.3 Product safety
 - 8.1.4 Prevention of counterfeit parts
- 8.2 Requirements for products and services
- 8.3 Design and development
- 8.4 Control of externally provided processes, products and services
 - 8.4.3 Information for external providers
- 8.5 Production and service provision
 - 8.5.1.2 Validation and control of special processes
 - 8.5.1.3 Production process verification (first article inspection)
- 8.5.2 Identification and traceability
- 8.6 Release of products and services
- 8.7 Control of nonconforming outputs

Clause 10: Improvement

- 10.1 General
- 10.2 Nonconformity and corrective action
- 10.3 Continual improvement

Clause 5: Leadership

- 5.1 Leadership and commitment
- 5.2 Policy
- 5.3 Organizational roles, responsibilities and authorities

Clause 7: Support

- 7.1 Resources
 - 7.1.5.2 Measurement traceability
- 7.2 Competence
- 7.3 Awareness
- 7.4 Communication
- 7.5 Documented information

Clause 9: Performance Evaluation

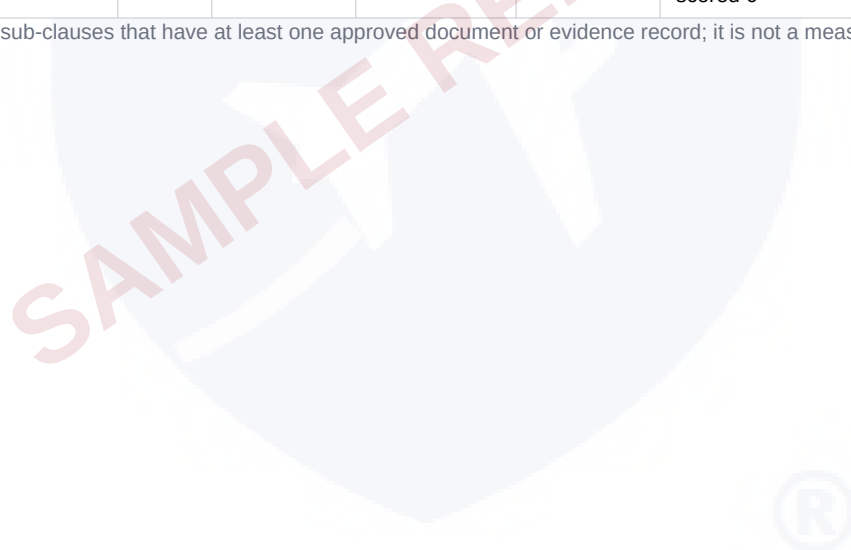
- 9.1 Monitoring, measurement, analysis and evaluation
- 9.2 Internal audit
- 9.3 Management review

3. Clause Compliance Posture

Aegis readiness score: a proprietary assessment of the documents, evidence and open findings recorded against each AS9100D clause. Not an audit result, and not a certification decision. Each clause is shown as Conforming, Minor gaps or Major risk, with what is on record against it and what would close each gap.

Clause	Title	Score	Status	Sub-clauses met	With evidence	Lowest gap
4	Context of the Organization	42.7	Major risk	0 of 4	4 of 4	4.3 Determining the scope of the quality management system: scored 10.8
5	Leadership	30.2	Major risk	0 of 3	3 of 3	5.2 Policy: scored 16.2
6	Planning	30.6	Major risk	0 of 3	3 of 3	6.2 Quality objectives and planning to achieve them: scored 27
7	Support	20.4	Major risk	0 of 5	5 of 5	7.5 Documented information: scored 5.5
8	Operation	10.5	Major risk	0 of 7	7 of 7	8.4 Control of externally provided processes, products, and services: scored 0
9	Performance Evaluation	28.3	Major risk	0 of 3	3 of 3	9.2 Internal audit: scored 21
10	Improvement	28.6	Major risk	0 of 3	3 of 3	10.2 Nonconformity and corrective action: scored 0

"With evidence" counts the sub-clauses that have at least one approved document or evidence record; it is not a measure of whether that evidence is sufficient.



Clause 4: Context of the Organization

Major risk, scored 42.7. Sub-clauses met: 0 of 4.

Sub-clause	Score	On record	To close it
4.1 Understanding the organization and its context	55	Approved: DOC-MER-013 14 audit or assessment evidence items	• Approve more documentation for 4.1
4.2 Understanding the needs and expectations of interested parties	44	No approved document 8 audit or assessment evidence items	• Issue an approved document for 4.2 (none on record) • Record more audit or assessment evidence against 4.2
4.3 Determining the scope of the quality management system	10.8	No approved document 1 nonconforming self-assessment answer 4 audit or assessment evidence items	• Resolve the nonconforming self-assessment answer • Issue an approved document for 4.3 (none on record) • Record more audit or assessment evidence against 4.3
4.4 Quality management system and its processes	55	Approved: DOC-MER-001 10 audit or assessment evidence items	• Approve more documentation for 4.4



Clause 5: Leadership

Major risk, scored 30.2. Sub-clauses met: 0 of 3.

Sub-clause	Score	On record	To close it
5.1 Leadership and commitment	27	No approved document 1 nonconforming self-assessment answer 18 audit or assessment evidence items	• Resolve the nonconforming self-assessment answer • Issue an approved document for 5.1 (none on record)
5.2 Policy	16.2	No approved document 1 nonconforming self-assessment answer 6 audit or assessment evidence items	• Resolve the nonconforming self-assessment answer • Issue an approved document for 5.2 (none on record) • Record more audit or assessment evidence against 5.2
5.3 Organizational roles, responsibilities and authorities	49	Approved: DOC-MER-014 8 audit or assessment evidence items	• Approve more documentation for 5.3 • Record more audit or assessment evidence against 5.3



Clause 6: Planning

Major risk, scored 30.6. Sub-clauses met: 0 of 3.

Sub-clause	Score	On record	To close it
6.1 Actions to address risks and opportunities	28	Approved: DOC-MER-015 2 nonconforming self-assessment answers 16 audit or assessment evidence items	• Resolve the 2 nonconforming self-assessment answers • Approve more documentation for 6.1
6.2 Quality objectives and planning to achieve them	27	No approved document 1 nonconforming self-assessment answer 10 audit or assessment evidence items	• Resolve the nonconforming self-assessment answer • Issue an approved document for 6.2 (none on record)
6.3 Planning of changes	38	No approved document 6 audit or assessment evidence items	• Issue an approved document for 6.3 (none on record) • Record more audit or assessment evidence against 6.3



Clause 7: Support

Major risk, scored 20.4. Sub-clauses met: 0 of 5.

Sub-clause	Score	On record	To close it
7.1 Resources	27	Approved: DOC-MER-016, DOC-MER-032, DOC-MER-039 Awaiting approval: DOC-MER-006 Open finding: FIND-MER-004 3 nonconforming self-assessment answers 20 audit or assessment evidence items	• Close FIND-MER-004 • Resolve the 3 nonconforming self-assessment answers • Approve DOC-MER-006 (awaiting approval) • Approve more documentation for 7.1
7.2 Competence	11.5	Approved: DOC-MER-017 Open finding: FIND-MER-005 4 nonconforming self-assessment answers 6 audit or assessment evidence items	• Close FIND-MER-005 • Resolve the 4 nonconforming self-assessment answers • Approve more documentation for 7.2 • Record more audit or assessment evidence against 7.2
7.3 Awareness	38	No approved document 6 audit or assessment evidence items	• Issue an approved document for 7.3 (none on record) • Record more audit or assessment evidence against 7.3
7.4 Communication	32	No approved document 4 audit or assessment evidence items	• Issue an approved document for 7.4 (none on record) • Record more audit or assessment evidence against 7.4
7.5 Documented information	5.5	Approved: DOC-MER-018 Open finding: FIND-MER-008 4 nonconforming self-assessment answers 2 audit or assessment evidence items	• Close FIND-MER-008 • Resolve the 4 nonconforming self-assessment answers • Approve more documentation for 7.5 • Record more audit or assessment evidence against 7.5

Clause 8: Operation

Major risk, scored 10.5. Sub-clauses met: 0 of 7.

Sub-clause	Score	On record	To close it
8.1 Operational planning and control	17.5	Approved: DOC-MER-019 5 nonconforming self-assessment answers 30 audit or assessment evidence items	- Resolve the 5 nonconforming self-assessment answers - Approve more documentation for 8.1
8.2 Requirements for products and services	16.2	No approved document 1 nonconforming self-assessment answer 6 audit or assessment evidence items	- Resolve the nonconforming self-assessment answer - Issue an approved document for 8.2 (none on record) - Record more audit or assessment evidence against 8.2
8.3 Design and development of products and services	37	Approved: DOC-MER-037 4 audit or assessment evidence items	- Approve more documentation for 8.3 - Record more audit or assessment evidence against 8.3
8.4 Control of externally provided processes, products, and services	0	Approved: DOC-MER-031 Awaiting approval: DOC-MER-020 Open finding: FIND-MER-007 10 nonconforming self-assessment answers 6 audit or assessment evidence items	- Close FIND-MER-007 - Resolve the 10 nonconforming self-assessment answers - Approve DOC-MER-020 (awaiting approval) - Approve more documentation for 8.4 - Record more audit or assessment evidence against 8.4
8.5 Production and service provision	0	Approved: DOC-MER-010, DOC-MER-021, DOC-MER-022, DOC-MER-029, DOC-MER-030, DOC-MER-033, DOC-MER-036, DOC-MER-038 Awaiting approval: DOC-MER-002, DOC-MER-009, DOC-MER-028 Open findings: FIND-MER-003, FIND-MER-011, FIND-MER-001, FIND-MER-006 and 15 nonconformances 25 nonconforming self-assessment answers 78 audit or assessment evidence items	- Close FIND-MER-003, FIND-MER-011, FIND-MER-001, FIND-MER-006 and the 15 open nonconformances - Resolve the 25 nonconforming self-assessment answers - Approve DOC-MER-002, DOC-MER-009, DOC-MER-028 (awaiting approval)
8.6 Release of products and services	0	Approved: DOC-MER-005, DOC-MER-035 Awaiting approval: DOC-MER-004 Open findings: FIND-MER-010 and 11 nonconformances 8 audit or assessment evidence items	- Close FIND-MER-010 and the 11 open nonconformances - Approve DOC-MER-004 (awaiting approval) - Approve more documentation for 8.6 - Record more audit or assessment evidence against 8.6
8.7 Control of nonconforming outputs	0	Approved: DOC-MER-007, DOC-MER-011, DOC-MER-023, DOC-MER-040 Open findings: 17 nonconformances 3 nonconforming self-assessment answers 6 audit or assessment evidence items	- Close the 17 open nonconformances - Resolve the 3 nonconforming self-assessment answers - Approve more documentation for 8.7 - Record more audit or assessment evidence against 8.7

Clause 9: Performance Evaluation

Major risk, scored 28.3. Sub-clauses met: 0 of 3.

Sub-clause	Score	On record	To close it
9.1 Monitoring, measurement, analysis and evaluation	28	Approved: DOC-MER-024 Awaiting approval: DOC-MER-012 2 nonconforming self-assessment answers 20 audit or assessment evidence items	• Resolve the 2 nonconforming self-assessment answers • Approve DOC-MER-012 (awaiting approval) • Approve more documentation for 9.1
9.2 Internal audit	21	Approved: DOC-MER-025 Open finding: FIND-MER-009 3 nonconforming self-assessment answers 10 audit or assessment evidence items	• Close FIND-MER-009 • Resolve the 3 nonconforming self-assessment answers • Approve more documentation for 9.2
9.3 Management review	36	Approved: DOC-MER-026, DOC-MER-034 1 nonconforming self-assessment answer 12 audit or assessment evidence items	• Resolve the nonconforming self-assessment answer • Approve more documentation for 9.3



Clause 10: Improvement

Major risk, scored 28.6. Sub-clauses met: 0 of 3.

Sub-clause	Score	On record	To close it
10.1 Improvement: general	50	No approved document 10 audit or assessment evidence items	• Issue an approved document for 10.1 (none on record)
10.2 Nonconformity and corrective action	0	Approved: DOC-MER-027 Open findings: 15 nonconformances 7 nonconforming self-assessment answers 4 audit or assessment evidence items	• Close the 15 open nonconformances • Resolve the 7 nonconforming self-assessment answers • Approve more documentation for 10.2 • Record more audit or assessment evidence against 10.2
10.3 Continual improvement	50	No approved document 12 audit or assessment evidence items	• Issue an approved document for 10.3 (none on record)



4. Evidence Index

Each controlled document and audit evidence record, with the AS9100D clause it supports. Documents that are awaiting approval, in draft or withdrawn are listed because an auditor will ask about them; only approved documents count as evidence in the clause scores.

Clause	Reference	Evidence	Type	Status	Owner
4.1	DOC-MER-013	QP-4.1 Context and Interested Parties	Procedure	Approved	G. Hoffman
4.4	DOC-MER-001	QMS-001 Quality Management System Manual	Procedure	Approved	V. Singh (QD)
5.3	DOC-MER-014	QP-5.3 Roles and Authorities	Procedure	Approved	V. Singh (QD)
6.1	DOC-MER-015	QP-6.1 Risk Management	Procedure	Approved	G. Hoffman
7.1.4	DOC-MER-039	QP-7.1.4 Work Environment and FOD	Procedure	Approved	P. Green
7.1.5	DOC-MER-006	CAL-500 Calibration Control Procedure	Procedure	Under review	L. Tran
7.1.5	DOC-MER-016	QP-7.1.5 Monitoring and Measuring Resources	Procedure	Approved	D. Reyes
7.1.5	DOC-MER-032	FRM-305 Calibration Recall Notice	Form	Approved	G. Hoffman
7.2	DOC-MER-008	TRN-700 Training Qualification & Records Management	Procedure	Obsolete	D. Kowalski
7.2	DOC-MER-017	QP-7.2 Competence	Procedure	Approved	S. Brooks
7.5	DOC-MER-018	QP-7.5 Control of Documented Information	Procedure	Approved	K. Osei
8.1.4	DOC-MER-019	QP-8.1.4 Counterfeit Parts Prevention	Procedure	Approved	K. Osei
8.3	DOC-MER-037	QP-8.3 Design Control (N/A statement)	Procedure	Approved	R. Vasquez
8.4	DOC-MER-020	QP-8.4 Control of External Providers	Procedure	Under review	T. Nakamura
8.4	DOC-MER-031	FRM-201 Supplier Scorecard	Form	Approved	M. Alvarez
8.5.1	DOC-MER-002	HT-100 Heat Treat Operation — Titanium Alloys	Work instruction	Under review	G. Hoffman
8.5.1	DOC-MER-003	PYR-200 Pyrometry Control per AMS 2750F	Procedure	Obsolete	G. Hoffman
8.5.1	DOC-MER-009	CNC-800 CNC Program Control & Verification	Work instruction	Under review	J. Reeves
8.5.1	DOC-MER-010	FAI-900 First Article Inspection Procedure per AS9102	Procedure	Approved	A. Chen
8.5.1	DOC-MER-028	WI-CNC-810 Tool Life Management	Work instruction	Under review	L. Tran
8.5.1	DOC-MER-029	WI-SM-120 Drill and Rivet	Work instruction	Approved	T. Nakamura
8.5.1	DOC-MER-030	WI-COMP-330 Layup and Bagging	Work instruction	Approved	G. Hoffman
8.5.1	DOC-MER-033	FRM-410 FAI Report Forms 1-3	Form	Approved	L. Tran
8.5.1	DOC-MER-036	WI-HT-110 Furnace Load Thermocouples	Work instruction	Approved	R. Vasquez
8.5.1.3	DOC-MER-038	QP-8.5.1.3 Production Process Verification	Procedure	Approved	D. Reyes
8.5.2	DOC-MER-021	QP-8.5.2 Identification and Traceability	Procedure	Approved	D. Reyes
8.5.6	DOC-MER-022	QP-8.5.6 Control of Changes	Procedure	Approved	M. Alvarez
8.6	DOC-MER-004	RI-300 Receiving Inspection — Raw Material	Work instruction	Under review	K. Williams

Clause	Reference	Evidence	Type	Status	Owner
8.6	DOC-MER-005	NDT-400 FPI Process Instruction per ASTM E1417	Work instruction	Approved	M. Okonkwo
8.6	DOC-MER-035	WI-NDT-420 UT Inspection	Work instruction	Approved	G. Hoffman
8.7	DOC-MER-007	MRB-600 Material Review Board Charter & Procedure	Procedure	Approved	V. Singh (QD)
8.7	DOC-MER-011	FRM-101 Nonconformance Report Form	Form	Approved	V. Singh (QD)
8.7	DOC-MER-023	QP-8.7 Control of Nonconforming Output	Procedure	Approved	D. Reyes
8.7	DOC-MER-040	FRM-150 Deviation Request	Form	Approved	L. Tran
9.1	DOC-MER-012	RPT-201 Monthly Quality Performance Report	Record	Under review	Recovery Lead
9.1.2	DOC-MER-024	QP-9.1.2 Customer Satisfaction	Procedure	Approved	S. Brooks
9.2	DOC-MER-025	QP-9.2 Internal Audit	Procedure	Approved	G. Hoffman
9.3	DOC-MER-026	QP-9.3 Management Review	Procedure	Approved	G. Hoffman
9.3	DOC-MER-034	RPT-220 Management Review Minutes	Record	Approved	T. Nakamura
10.2	DOC-MER-027	QP-10.2 Nonconformity and Corrective Action	Procedure	Approved	D. Reyes

Proof on File for Closed Records

The documents behind your records stay in your systems; the registers say where each is kept. Of 126 closed or held records, 115 name where the proof is kept (CAPA 43 of 46; FAI 2 of 3; MRB 27 of 30; Training 43 of 47). An auditor samples closed records and asks to see that proof, so before the audit pull a sample of the referenced documents and confirm each exists and says what the register records.

11 records closed or held with nowhere named for the proof. Find each before the audit and add its reference; the same list is 12_Evidence_Requests in the pack.

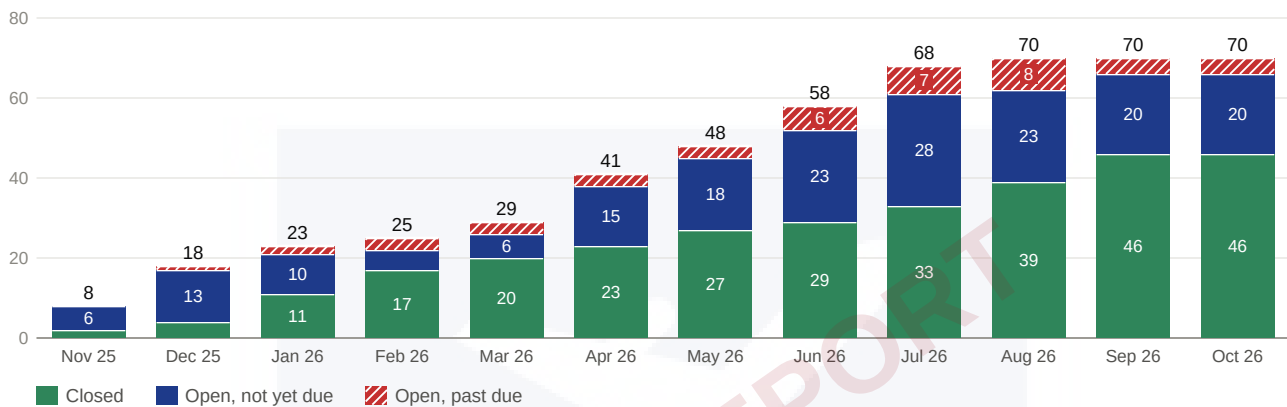
Register	Clause	Record	What	State	Owner
CAPA	10.2	CAPA-MER-022	Porosity at FPI — MER-3118	Closed	D. Reyes
CAPA	10.2	CAPA-MER-045	Cure cycle deviation — MER-3102	Closed	L. Tran
CAPA	10.2	CAPA-MER-060	Bore oversize — MER-3113	Closed	S. Brooks
FAI	8.5.1.3	FAI-MER-105	MER-3105 Floor Beam Support Clip	Approved	Not recorded
MRB	8.7	MRB-MER-313	Hardness below spec after heat treat — MER-3117	Closed	G. Hoffman
MRB	8.7	MRB-MER-324	Missing inspection stamp — MER-3103	Closed	V. Singh (QD)
MRB	8.7	MRB-MER-332	Delamination — MER-3108	Closed	V. Singh (QD)
Training	7.2	MER-E014	Taylor Haddad — AMS Material ID & CoC Verification	Held	Taylor Haddad
Training	7.2	MER-E039	Jamie Okafor — AS9102 FAI Documentation	Held	Jamie Okafor
Training	7.2	MER-E056	Kendall Lindqvist — Autoclave Cure Cycle Qualification	Held	Kendall Lindqvist
Training	7.2	MER-E067	Skyler Sato — AMS Material ID & CoC Verification	Held	Skyler Sato

5. Open Issues Package

Every open corrective action, audit finding and risk, with who holds it and when it is due. An item with no owner is shown as not assigned rather than left blank.

5.1 Open CAPAs

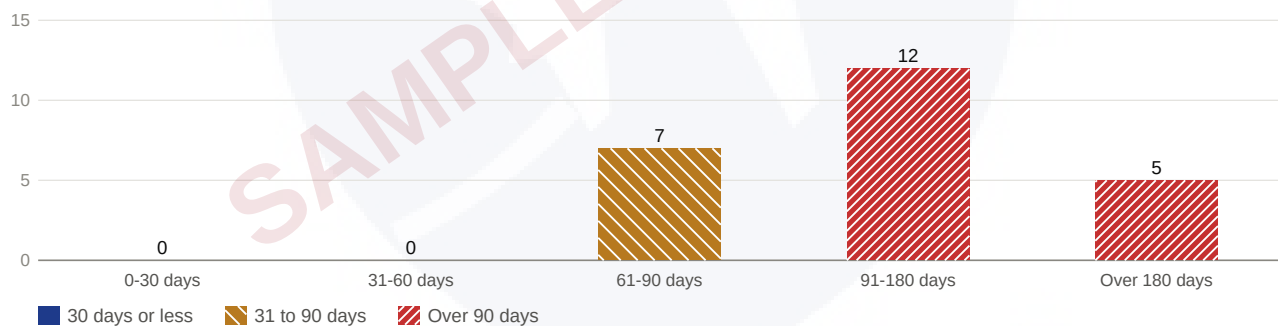
CAPAs month by month



The CAPA register at each month end, from its opened, closed and due dates; the last month ends on the report date.

CAPAs at Oct 26 (report date): 46 closed, 20 open and not yet due, 4 past due; at Nov 25: 2 closed, 6 open, 0 past due.

Open CAPAs by age



Open CAPAs by days since opened.

24 open CAPAs by age: 0 0-30 days, 0 31-60 days, 7 61-90 days, 12 91-180 days, 5 Over 180 days.

24 open, 4 past their due date; 46 closed (37 with effectiveness verified, 9 awaiting the check); 60.9% of closures were on or before the due date.

CAPA	NCR	Title	Root cause	Effectiveness	Status	Owner	Due	Days past due
CAPA-MER-029	NCR-MER-018	Missing inspection stamp — MER-3105	Final inspection signed in batches at shift end	Pending — past due, re-plan not agreed	Implementation	T. Nakamura	2025-12-25	287

CAPA	NCR	Title	Root cause	Effectiveness	Status	Owner	Due	Days past due
CAPA-MER-042	NCR-MER-031	Wrong revision program run — MER-3110	Programs pulled from a shared folder without revision check	Pending — past due, re-plan not agreed	Implementation	R. Vasquez	2026-03-01	221
CAPA-MER-052	NCR-MER-041	Bore oversize — MER-3107	Boring bar replaced on hours, not on measured wear; tool life exceeded	Pending — past due, re-plan not agreed	Implementation	P. Green	2026-07-08	92
CAPA-MER-019	NCR-MER-005	Edge distance violation — MER-3109	Not recorded	Pending — past due, re-plan not agreed	Root cause	R. Vasquez	2026-07-11	89
CAPA-MER-007	NCR-MER-207	Calibration overdue — 32 assets including CMM and hardness testers	Calibration recall run from a spreadsheet nobody owned during the QM vacancy; no escalation when an asset passed its due date	Pending — vendor scheduling started	Implementation	L. Tran	2026-10-23	-
CAPA-MER-002	NCR-MER-202	Pyrometry gap — 6-month hole in heat treat furnace #3 records	Not recorded	Pending — containment only	Initiated	G. Hoffman	2026-10-25	-
CAPA-MER-006	NCR-MER-206	CNC program revision control — obsolete G-code run on 5-axis center	CNC programs released from a shared folder with no revision lock; the controller ran the last file saved	Pending — interim manual verification in place	Action planning	J. Reeves	2026-10-27	-
CAPA-MER-008	NCR-MER-208	Expired operator certifications — 12 machinists and inspectors	Certification expiry not tracked against the shift roster; the training coordinator role was vacant for five months	Pending — training schedule drafted	Action planning	D. Kowalski	2026-10-29	-
CAPA-MER-024	NCR-MER-012	Porosity at FPI — MER-3103	Casting porosity from the supplier's degassing step, not screened at receipt	Pending — actions in progress	Action planning	D. Reyes	2026-10-29	-

CAPA	NCR	Title	Root cause	Effectiveness	Status	Owner	Due	Days past due
CAPA-MER-031	NCR-MER-020	Cure cycle deviation — MER-3102	Thermocouple lead pinched in the bag; cure controlled on a failed reading	Pending — actions in progress	Implementation	L. Tran	2026-10-29	-
CAPA-MER-032	NCR-MER-021	Heat lot mismatch on CoC — MER-3107	Receiving checked the CoC for presence, not heat lot against the mill cert	Pending — actions in progress	Implementation	T. Nakamura	2026-10-29	-
CAPA-MER-040	NCR-MER-029	Missing inspection stamp — MER-3103	Final inspection signed in batches at shift end	Pending — actions in progress	Implementation	T. Nakamura	2026-10-29	-
CAPA-MER-070	NCR-MER-060	Bore oversize — MER-3114	Boring bar replaced on hours, not on measured wear; tool life exceeded	Pending — actions in progress	Implementation	K. Osei	2026-10-29	-
CAPA-MER-001	NCR-MER-201	Ti-6Al-4V forging inclusion — wrong heat lot traceability	Not recorded	Pending — no root cause started	Initiated	R. Vasquez	2026-10-30	-
CAPA-MER-004	NCR-MER-204	NDT FPI false-call rate 14% on aluminum wing skins	FPI line developer dwell set by operator judgement, not timed; two Level I operators reading indications unsupervised	Pending — operator requalification started	Implementation	M. Okonkwo	2026-11-02	-
CAPA-MER-003	NCR-MER-203	A320neo nacelle bracket OTD failure — 4 consecutive months <60%	Not recorded	Pending — no action plan	Root cause	T. Nakamura	2026-11-04	-
CAPA-MER-005	NCR-MER-205	Composite layup cure cycle deviation — 3 autoclave batches out of spec	Not recorded	Pending — thermocouple investigation underway	Root cause	S. Petrov	2026-11-06	-

CAPA	NCR	Title	Root cause	Effectiveness	Status	Owner	Due	Days past due
CAPA-MER-009	NCR-MER-209	Boeing rivet spacing nonconformance — fuselage panel lot 26-018	Not recorded	Pending — lot quarantined	Root cause	A. Chen	2026-11-09	-
CAPA-MER-014	NCR-MER-214	GE Aviation engine mount bracket hardness out of spec	Not recorded	Pending — heat treat parameters under review	Root cause	R. Vasquez	2026-11-12	-
CAPA-MER-010	NCR-MER-210	Chemical processing tank concentration drift — chromate conversion	Titration frequency cut from daily to weekly to save lab time; drift went undetected between checks	Pending — tank drained and recharged	Verification	P. Gutierrez	2026-11-15	-
CAPA-MER-011	NCR-MER-211	Supplier raw material CoC discrepancies — 3 titanium lots	Not recorded	Pending — supplier notified	Root cause	K. Williams	2026-11-19	-
CAPA-MER-013	NCR-MER-213	FOD control failure in sheet metal shop — debris in assembly	No FOD walk at shift change in the sheet metal shop; drill swarf left on benches feeding assembly	Pending — clean-as-you-go audit failed	Implementation	B. Harper	2026-11-22	-
CAPA-MER-012	NCR-MER-212	Prevent recurrence of quality manager vacancy gap	No deputy named for the quality manager; the role's duties were not written down	Pending — succession plan drafted	Action planning	V. Singh (QD)	2026-11-24	-
CAPA-MER-015	NCR-MER-215	Prevent audit finding recurrence — establish internal audit cadence	Internal audit schedule held only in the former quality manager's calendar; nobody else could run it	Pending — schedule proposed	Action planning	V. Singh (QD)	2026-11-29	-

5.2 Open Audit Findings

Finding	Description	Severity	Clause	Status	Assigned to	Due
FIND-MER-001	Ti-6Al-4V traceability failure — 3 forging lots with incorrect heat lot on CoC	Major	AS9100D 8.5.2	Open	R. Vasquez	2026-10-30
FIND-MER-002	Pyrometry records — 6-month gap in furnace #3 survey documentation	Major	AMS 2750F §3.4	Open	G. Hoffman	2026-10-25
FIND-MER-003	Systematic OTD failure — 4 consecutive months below 60% on nacelle bracket program	Major	AS9100D 8.5.1	Open	T. Nakamura	2026-11-04
FIND-MER-004	32 calibration assets overdue — measurement system integrity compromised	Major	AS9100D 7.1.5	Open	L. Tran	2026-10-23
FIND-MER-005	12 expired operator certifications on active production lines	Major	AS9100D 7.2	Open	D. Kowalski	2026-10-29
FIND-MER-006	Rivet spacing nonconformance on fuselage panel lot 26-018 — acceptance criteria not met	Major	AS9100D 8.5.1	Open	A. Chen	2026-11-09
FIND-MER-007	Supplier CoC discrepancies — 3 titanium lots from Great Plains Titanium with formatting errors	Minor	AS9100D 8.4.2	Open	K. Williams	2026-11-19
FIND-MER-008	Pyrometry control procedure (PYR-200) obsolete — operators working without valid WI	Minor	AS9100D 7.5	Open	G. Hoffman	2026-10-27
FIND-MER-009	Internal audit program lapsed — no internal audits for 4 months (May to September) during QM vacancy	Minor	AS9100D 9.2	Open	V. Singh (QD)	2026-11-29
FIND-MER-010	Engine mount bracket hardness below spec — product released without proper disposition	Minor	AS9100D 8.6	Open	R. Vasquez	2026-11-12
FIND-MER-011	CNC program revision control failure — obsolete G-code executed on 5-axis center	Minor	AS9100D 8.5.1	Open	J. Reeves	2026-10-27
FIND-MER-012	Radiographic inspection film density outside acceptable range on 2 lots	Minor	ASTM E2104	Open	Not assigned	2026-11-14

5.3 Open Risks

Risk	Description	Score	Status	Owner	Due	Mitigation
RISK-MER-01	CB surveillance audit failure leading to AS9100D suspension	25	Identified	V. Singh (QD)	2026-11-04	Pre-audit readiness blitz — gap closure plan in 14 days
RISK-MER-02	Airbus de-source action on A320neo nacelle bracket program	20	Identified	T. Nakamura	2026-11-09	Weekly recovery brief + dedicated expediting cell
RISK-MER-03	NADCAP heat treatment accreditation revoked	20	Identified	G. Hoffman	2026-10-25	Pyrometry record restoration + furnace requalification
RISK-MER-04	Boeing contract termination for chronic SCAR non-closure	20	Identified	R. Vasquez	2026-11-02	SCAR response team + root cause sprint

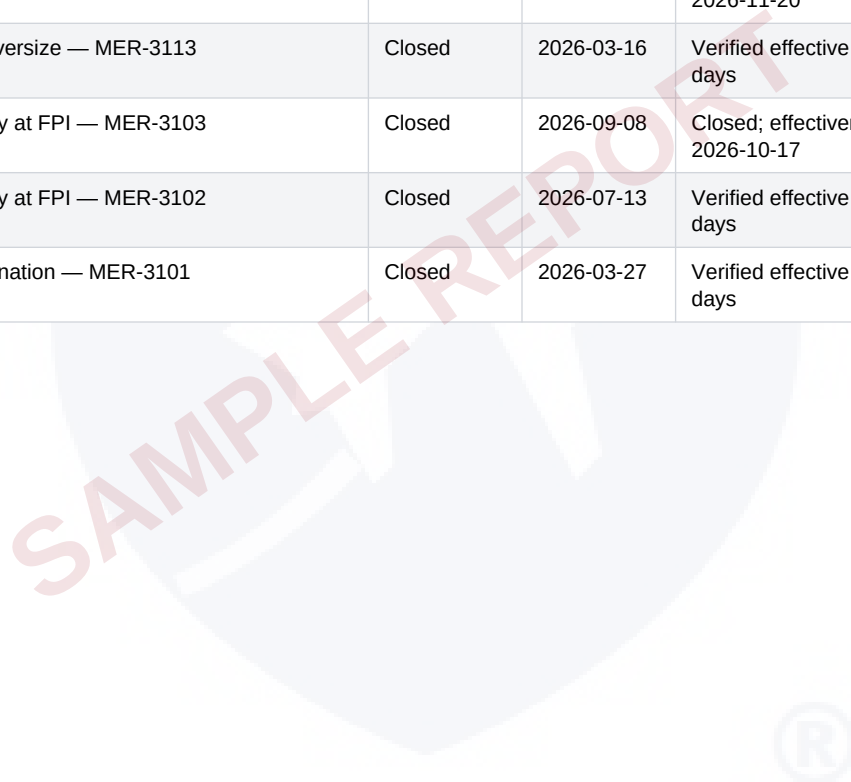
Risk	Description	Score	Status	Owner	Due	Mitigation
RISK-MER-06	Measurement system integrity — 32 overdue calibrations invalidate acceptance data	20	Identified	L. Tran	2026-10-23	Emergency calibration vendor mobilization
RISK-MER-05	Titanium raw material traceability gaps expose latent field escapes	16	Identified	K. Williams	2026-11-06	Lot-by-lot containment review + supplier CoC reconciliation
RISK-MER-07	Workforce competency — 12 expired operator certs on active production lines	16	Identified	D. Kowalski	2026-10-29	Accelerated requalification program + interim supervision
RISK-MER-09	GE Aviation engine mount bracket field return — hardness failure	15	Identified	R. Vasquez	2026-11-12	Lot containment + heat treat parameter revalidation
RISK-MER-08	Spirit AeroSystems quality escalation — second-tier customer ripple effect	12	Identified	T. Nakamura	2026-11-15	Proactive communication + shared corrective action visibility
RISK-MER-10	Insurance/liability exposure from uncontrolled product release	12	Identified	V. Singh (QD)	2026-11-24	Legal review + product containment verification

5.4 Closed CAPAs (effectiveness checks)

CAPA	Title	Status	Closed	Effectiveness criteria
CAPA-MER-016	Hardness below spec after heat treat — MER-3117	Closed	2026-09-20	Closed; effectiveness check due 2026-12-04
CAPA-MER-017	Hardness below spec after heat treat — MER-3106	Closed	2026-04-23	Verified effective — no recurrence in 45 days
CAPA-MER-018	Heat lot mismatch on CoC — MER-3114	Closed	2026-02-15	Verified effective — no recurrence in 73 days
CAPA-MER-020	Bore oversize — MER-3102	Closed	2026-01-15	Verified effective — no recurrence in 74 days
CAPA-MER-021	Surface finish Ra out of spec — MER-3114	Closed	2026-02-10	Verified effective — no recurrence in 50 days
CAPA-MER-022	Porosity at FPI — MER-3118	Closed	2026-01-17	Verified effective — no recurrence in 50 days
CAPA-MER-023	Delamination — MER-3102	Closed	2026-08-06	Verified effective — no recurrence in 49 days
CAPA-MER-025	Edge distance violation — MER-3106	Closed	2026-07-08	Verified effective — no recurrence in 68 days
CAPA-MER-026	Edge distance violation — MER-3102	Closed	2026-08-11	Verified effective — no recurrence in 41 days
CAPA-MER-027	Heat lot mismatch on CoC — MER-3107	Closed	2026-08-06	Verified effective — no recurrence in 35 days
CAPA-MER-028	Porosity at FPI — MER-3115	Closed	2026-01-17	Verified effective — no recurrence in 74 days
CAPA-MER-030	Missing inspection stamp — MER-3115	Closed	2025-11-18	Verified effective — no recurrence in 61 days

CAPA	Title	Status	Closed	Effectiveness criteria
CAPA-MER-033	Wrong revision program run — MER-3112	Closed	2026-09-03	Closed; effectiveness check due 2026-11-08
CAPA-MER-034	Hardness below spec after heat treat — MER-3104	Closed	2026-01-18	Verified effective — no recurrence in 64 days
CAPA-MER-035	Porosity at FPI — MER-3105	Closed	2026-07-09	Verified effective — no recurrence in 72 days
CAPA-MER-036	Edge distance violation — MER-3112	Closed	2026-08-06	Verified effective — no recurrence in 43 days
CAPA-MER-037	Edge distance violation — MER-3117	Closed	2026-09-16	Closed; effectiveness check due 2026-11-06
CAPA-MER-038	Cure cycle deviation — MER-3110	Closed	2026-05-20	Verified effective — no recurrence in 48 days
CAPA-MER-039	Porosity at FPI — MER-3109	Closed	2026-06-12	Verified effective — no recurrence in 67 days
CAPA-MER-041	Porosity at FPI — MER-3102	Closed	2026-05-23	Verified effective — no recurrence in 72 days
CAPA-MER-043	Cure cycle deviation — MER-3112	Closed	2025-11-28	Verified effective — no recurrence in 52 days
CAPA-MER-044	Surface finish Ra out of spec — MER-3117	Closed	2026-04-29	Verified effective — no recurrence in 59 days
CAPA-MER-045	Cure cycle deviation — MER-3102	Closed	2026-02-03	Verified effective — no recurrence in 38 days
CAPA-MER-046	Delamination — MER-3109	Closed	2026-02-01	Verified effective — no recurrence in 47 days
CAPA-MER-047	Delamination — MER-3108	Closed	2026-04-19	Verified effective — no recurrence in 33 days
CAPA-MER-048	Delamination — MER-3110	Closed	2025-12-04	Verified effective — no recurrence in 36 days
CAPA-MER-049	Edge distance violation — MER-3116	Closed	2026-08-29	Closed; effectiveness check due 2026-10-01
CAPA-MER-050	Bore oversize — MER-3113	Closed	2026-09-13	Closed; effectiveness check due 2026-11-27
CAPA-MER-051	Porosity at FPI — MER-3116	Closed	2026-06-16	Verified effective — no recurrence in 52 days
CAPA-MER-053	Bore oversize — MER-3109	Closed	2026-03-29	Verified effective — no recurrence in 65 days
CAPA-MER-054	Delamination — MER-3103	Closed	2026-08-08	Closed; effectiveness check due 2026-10-13
CAPA-MER-055	Porosity at FPI — MER-3113	Closed	2026-09-18	Closed; effectiveness check due 2026-10-24
CAPA-MER-056	Porosity at FPI — MER-3118	Closed	2026-02-08	Verified effective — no recurrence in 35 days
CAPA-MER-057	Delamination — MER-3108	Closed	2026-07-29	Verified effective — no recurrence in 40 days
CAPA-MER-058	Surface finish Ra out of spec — MER-3103	Closed	2026-01-07	Verified effective — no recurrence in 66 days

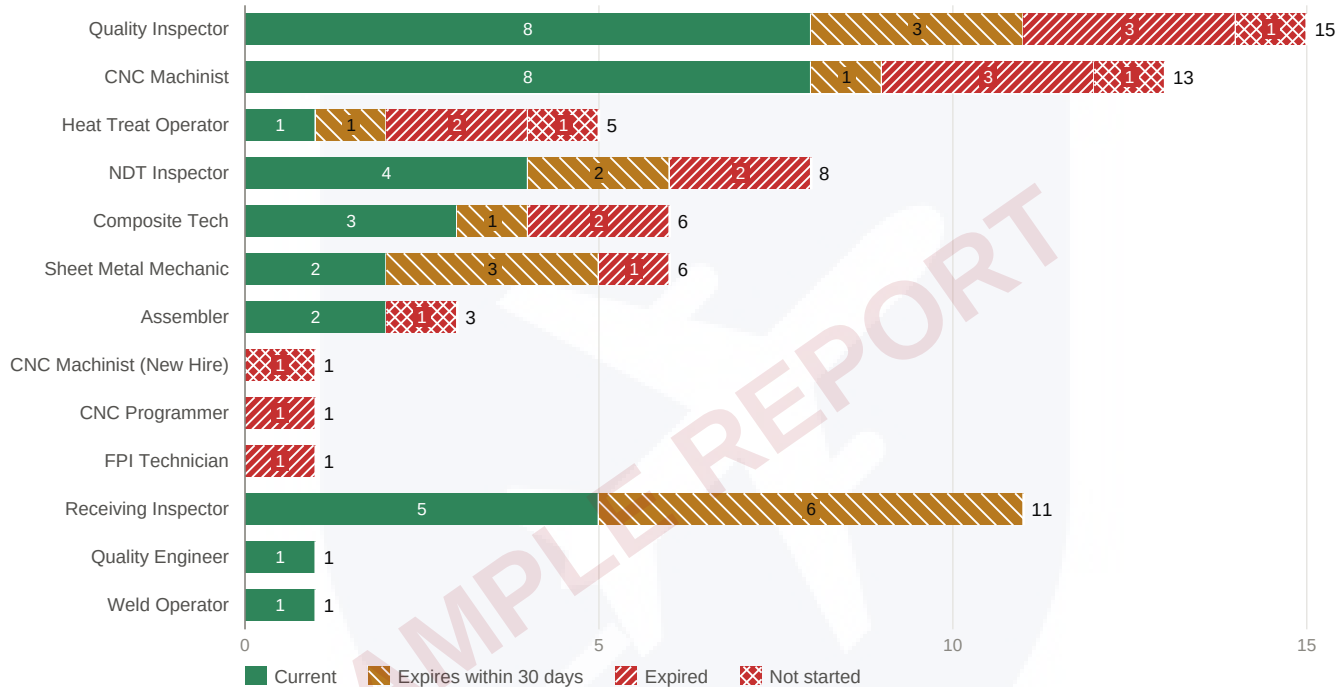
CAPA	Title	Status	Closed	Effectiveness criteria
CAPA-MER-059	Wrong revision program run — MER-3115	Closed	2026-05-10	Verified effective — no recurrence in 55 days
CAPA-MER-060	Bore oversize — MER-3113	Closed	2026-05-22	Verified effective — no recurrence in 36 days
CAPA-MER-061	Bore oversize — MER-3103	Closed	2026-01-21	Verified effective — no recurrence in 72 days
CAPA-MER-062	Wrong revision program run — MER-3101	Closed	2025-12-04	Verified effective — no recurrence in 32 days
CAPA-MER-063	Missing inspection stamp — MER-3110	Closed	2026-02-18	Verified effective — no recurrence in 36 days
CAPA-MER-064	Delamination — MER-3107	Closed	2026-01-02	Verified effective — no recurrence in 65 days
CAPA-MER-065	Surface finish Ra out of spec — MER-3111	Closed	2026-09-13	Closed; effectiveness check due 2026-11-20
CAPA-MER-066	Bore oversize — MER-3113	Closed	2026-03-16	Verified effective — no recurrence in 30 days
CAPA-MER-067	Porosity at FPI — MER-3103	Closed	2026-09-08	Closed; effectiveness check due 2026-10-17
CAPA-MER-068	Porosity at FPI — MER-3102	Closed	2026-07-13	Verified effective — no recurrence in 30 days
CAPA-MER-069	Delamination — MER-3101	Closed	2026-03-27	Verified effective — no recurrence in 62 days



6. Training & Competency Snapshot

Measure	Value	What it measures
Training assignment completion (%)	65.3	47 of 72 training assignments completed.
Expired qualifications	15	Qualification records whose expiry date has passed.
People with an expired qualification	15	Distinct people holding at least one expired qualification.

Qualification currency by role



Qualifications held, by the holder's role in the training matrix, on the report date. Not started: assigned and not yet begun.

Quality Inspector: 8 current, 3 expires within 30 days, 3 expired, 1 not started; CNC Machinist: 8 current, 1 expires within 30 days, 3 expired, 1 not started; Heat Treat Operator: 1 current, 1 expires within 30 days, 2 expired, 1 not started; NDT Inspector: 4 current, 2 expires within 30 days, 2 expired; Composite Tech: 3 current, 1 expires within 30 days, 2 expired; Sheet Metal Mechanic: 2 current, 3 expires within 30 days, 1 expired; Assembler: 2 current, 1 not started; CNC Machinist (New Hire): 1 not started; CNC Programmer: 1 expired; FPI Technician: 1 expired; Receiving Inspector: 5 current, 6 expires within 30 days; Quality Engineer: 1 current; Weld Operator: 1 current.

Expired qualifications

Employee	Name	Qualification	Expired
MER-E006	Aisha Patel	AS9102 FAI Documentation	2026-05-01
MER-E022	Sawyer Mbeki	GD&T per ASME Y14.5	2026-06-08
MER-E002	Janice Whitfield	ASNT NDT Level II (UT)	2026-06-16
MER-E004	Maria Santos	Autoclave Cure Cycle Qualification	2026-06-17
MER-E001	Carlos Rivera	NADCAP Heat Treat Operator Qual	2026-07-01
MER-E025	Parker Castillo	AMS 2750 Pyrometry Requirements	2026-07-05

Employee	Name	Qualification	Expired
MER-E031	Parker Chen	GD&T per ASME Y14.5	2026-07-20
MER-E003	Derek Fung	Fanuc CNC Programming Cert	2026-07-27
MER-E021	Riley Yamada	ASNT NDT Level II (PT)	2026-08-10
MER-E005	Kevin O'Brien	Boeing BAC 5000 Series Awareness	2026-08-14
MER-E012	Samantha Brooks	ASNT NDT Level II (PT/MT)	2026-08-18
MER-E030	Reese Castillo	AS9102 FAI Documentation	2026-08-26
MER-E049	Drew Molina	Autoclave Cure Cycle Qualification	2026-10-03
MER-E058	Jamie Haddad	CMM Operation (Calypso)	2026-10-06
MER-E013	Alex Haddad	AMS 2750 Pyrometry Requirements	2026-10-07

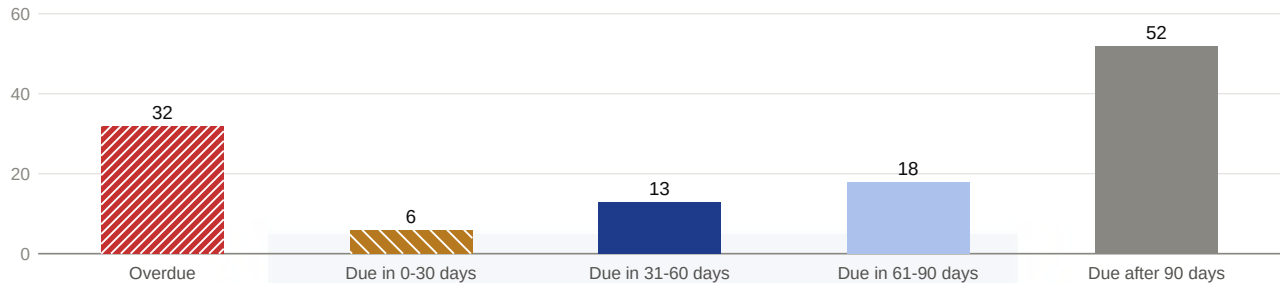
SAMPLE REPORT



7. Calibration & MSA

Calibration Compliance: 73.6% · Past due: 32 · Failed calibration: 3

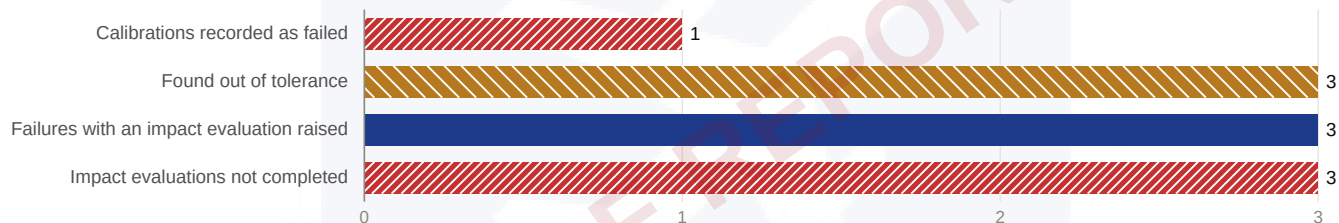
When each gauge falls due



Measuring equipment in service by when its next calibration falls due from the report date. Overdue is as your calibration register records it.

121 gauges in service: 32 overdue, 6 due in 0-30 days, 13 due in 31-60 days, 18 due in 61-90 days, 52 due after 90 days.

Failed and out-of-tolerance calibrations



From 111 calibrations recorded in the 12 months to the report date, and the impact evaluations raised for failures.

111 calibrations in 12 months: 1 failed, 3 found out of tolerance; 3 impact evaluations raised, 3 not completed.

Failed calibration and product impact (7.1.5.2)

Asset	Equipment	Failed	Evaluation	In doubt	Disposition
CAL-MER-121	Bore Gauge 12mm (cell 4)	2026-09-19	Pending evaluation	Not yet assessed	Not yet dispositioned
CAL-MER-02	Wilson Rockwell Hardness Tester	2026-09-21	Pending evaluation	Not yet assessed	Not yet dispositioned
CAL-MER-03	Fluke 1586A Super-DAQ Pyrometer	2026-09-21	Pending evaluation	Not yet assessed	Not yet dispositioned

"Not yet assessed" means no measurement record links this equipment to the parts it accepted. Supply the gauge-usage records (instrument, work order, date) to size the suspect population; until then it is unbounded, not zero.

Overdue calibrations

Risk Statement: Overdue calibrations may impact product acceptance decisions made using affected measurement equipment.

Asset	Serial	Equipment	Location	Due	Days past due
CAL-MER-018	TPG-7018	Thread Plug Gauge Set	Receiving	2026-04-21	170
CAL-MER-015	MIC-7015	Mitutoyo Outside Micrometer 0-25mm	CNC Shop Floor	2026-04-26	165

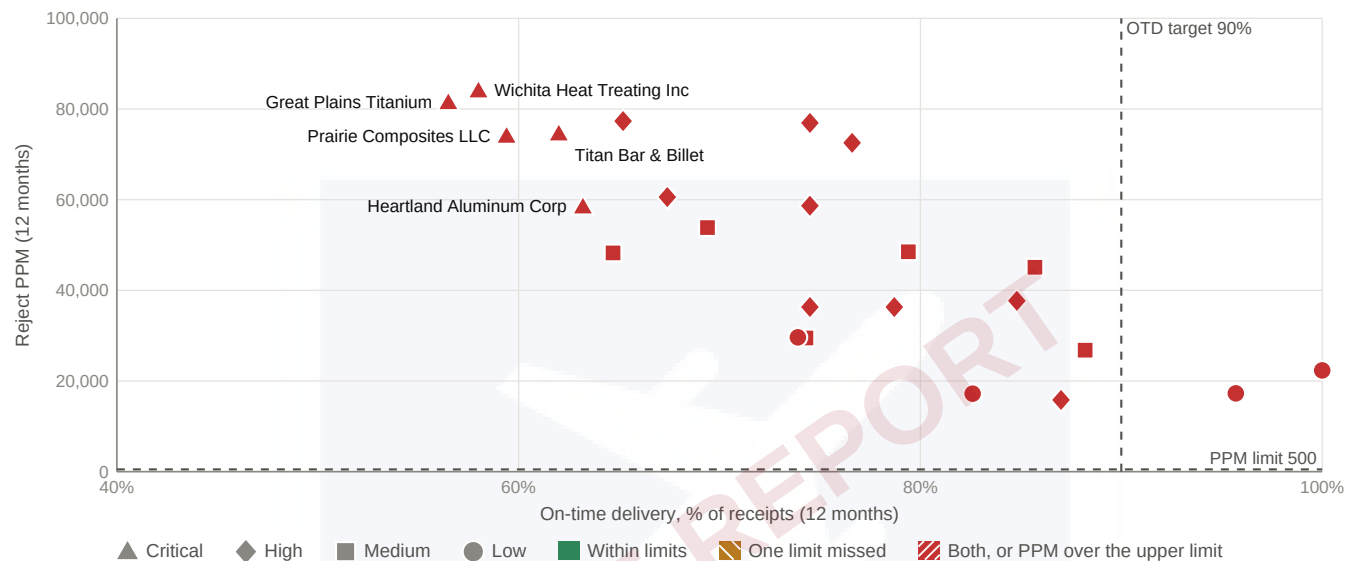
Asset	Serial	Equipment	Location	Due	Days past due
CAL-MER-022	PRF-7022	Profilometer SJ-210	Inspection Bay 2	2026-05-07	154
CAL-MER-021	PG-7021	Pressure Gauge 0-100 psi	Autoclave	2026-05-09	152
CAL-MER-014	TW-7014	Torque Wrench 20-100 Nm	Sub-Assembly	2026-05-15	146
CAL-MER-029	MIC-7029	Mitutoyo Outside Micrometer 0-25mm	CNC Shop Floor	2026-05-16	145
CAL-MER-035	TPG-7035	Thread Plug Gauge Set	Receiving	2026-05-24	137
CAL-MER-024	TPG-7024	Thread Plug Gauge Set	Receiving	2026-06-10	120
CAL-MER-031	TW-7031	Torque Wrench 20-100 Nm	Sub-Assembly	2026-06-14	116
CAL-MER-016	MIC-7016	Mitutoyo Outside Micrometer 0-25mm	CNC Shop Floor	2026-06-25	105
CAL-MER-01	CMM-4401	Zeiss Contura CMM	Inspection Bay 1	2026-07-01	99
CAL-MER-013	CAL-7013	Mitutoyo Digital Caliper 150mm	CNC Shop Floor	2026-07-03	97
CAL-MER-033	MIC-7033	Mitutoyo Outside Micrometer 0-25mm	CNC Shop Floor	2026-07-12	88
CAL-MER-012	SP-7012	Surface Plate Grade A	Inspection Bay 1	2026-07-15	85
CAL-MER-08	SHG-5503	Starrett Height Gauge 0-600mm	Sheet Metal Shop	2026-07-16	84
CAL-MER-121	BG-14	Bore Gauge 12mm (cell 4)	CNC Cell 4	2026-07-16	84
CAL-MER-023	MIC-7023	Mitutoyo Outside Micrometer 0-25mm	CNC Shop Floor	2026-07-17	83
CAL-MER-030	CAL-7030	Mitutoyo Digital Caliper 150mm	CNC Shop Floor	2026-07-21	79
CAL-MER-027	MIC-7027	Mitutoyo Outside Micrometer 0-25mm	CNC Shop Floor	2026-07-26	74
CAL-MER-011	MIC-7011	Mitutoyo Outside Micrometer 0-25mm	CNC Shop Floor	2026-07-28	72
CAL-MER-025	BG-7025	Bore Gauge 10-18mm	CNC Shop Floor	2026-08-07	62
CAL-MER-034	BG-7034	Bore Gauge 10-18mm	CNC Shop Floor	2026-08-10	59
CAL-MER-04	UT-3310	Olympus OmniScan MX2 UT System	NDT Cell	2026-08-18	51
CAL-MER-028	DUR-7028	Durometer Shore A	Receiving	2026-08-20	49
CAL-MER-036	CAL-7036	Mitutoyo Digital Caliper 150mm	CNC Shop Floor	2026-08-23	46
CAL-MER-019	TPG-7019	Thread Plug Gauge Set	Receiving	2026-08-25	44
CAL-MER-032	DUR-7032	Durometer Shore A	Receiving	2026-08-25	44
CAL-MER-020	TC-7020	Type K Thermocouple	Furnace #2	2026-08-27	42
CAL-MER-09	TC-9912	Autoclave Thermocouple Array (12 ch)	Composite Bay	2026-09-08	30
CAL-MER-017	PRF-7017	Profilometer SJ-210	Inspection Bay 2	2026-09-08	30
CAL-MER-026	TPG-7026	Thread Plug Gauge Set	Receiving	2026-09-18	20
CAL-MER-05	BG-6601	Mitutoyo Bore Gauge Set 50-150mm	CNC Shop Floor	2026-09-26	12

Total equipment tracked: 121

8. Supplier Governance

25 suppliers on record; 12 need attention. Each row joins the supplier's scores with its special-process accreditation (8.5.1.2), its sub-tiers' NADCAP accreditation (8.4.2), requirement flow-down (8.4.3), counterfeit cases (8.1.4) and risk index.

Delivery against rejects



Each supplier with receipts: delivery and rejects from its receipts. Marker shape is the criticality on the supplier list; the furthest out are named.

24 suppliers with receipts, 24 missing a limit. Furthest out: Great Plains Titanium 56.5% on time, 81,391 PPM (critical); Wichita Heat Treating Inc 58% on time, 83,974 PPM (critical); Prairie Composites LLC 59.4% on time, 73,973 PPM (critical); Titan Bar & Billet 62% on time, 74,483 PPM (critical); Flint Chemical Processing 65.2% on time, 77,333 PPM (high); Heartland Aluminum Corp 63.2% on time, 58,378 PPM (critical).

Supplier	Status	Criticality	OTD %	Quality	Reject PPM	Risk index	Flow-down	Special proc. without current approval	Sub-tier NADCAP gaps	Counterfeit open
Great Plains Titanium	Probation	Critical	56.5	68	81,391	72 (Medium)	0/1 flowed (0 verified)	-	1	1
Heartland Aluminum Corp	Probation	Critical	63.2	74	58,378	38.8 (Low)	none mapped	-	1	-
Aero Fastener Brokers	Conditional	High	74.5	69	76,923	20.2 (Low)	0/1 flowed, 1 partly (0 verified)	-	-	2
Great Bend Machining	Conditional	Medium	64.7	81	48,254	18 (Low)	0/1 flowed (0 verified)	-	1	-
Prairie Composites LLC	Probation	Critical	59.4	71	73,973	55.7 (Medium)	none mapped	-	-	-
Wichita Heat Treating Inc	Conditional	Critical	58	65	83,974	45.6 (Medium)	1/1 flowed (1 verified)	-	-	-
Titan Bar & Billet	Probation	Critical	62	70	74,483	39.8 (Low)	0/1 flowed (0 verified)	-	-	-

Supplier	Status	Criticality	OTD %	Quality	Reject PPM	Risk index	Flow-down	Special proc. without current approval	Sub-tier NADCAP gaps	Counterfeit open
Emporia Anodizing Co	Probation	High	76.6	71	72,537	28.5 (Low)	0/1 flowed (0 verified)	-	-	-
Flint Chemical Processing	Conditional	High	65.2	72	77,333	23.1 (Low)	none mapped	-	-	-
Kansas Precision Castings	Conditional	High	67.4	76	60,571	22 (Low)	none mapped	-	-	-
Central NDT Services	Conditional	High	74.5	78	58,659	19.8 (Low)	none mapped	-	-	-
Salina Sheet Metal Works	Conditional	Medium	69.4	79	53,846	17 (Low)	none mapped	-	-	-
Manhattan Metrology Services	Approved	High	74.5	86	36,316	8.9 (Low)	none mapped	-	-	-
Midwest Fastener Supply	Approved	High	78.7	83	36,311	7.5 (Low)	none mapped	-	-	-
Plains Surface Technologies	Approved	Medium	79.4	80	48,507	7.1 (Low)	none mapped	-	-	-
Flint Hills Adhesives	Approved	Medium	74.3	87	29,493	7 (Low)	none mapped	-	-	-
Precision Tooling Solutions	Approved	High	84.8	86	37,688	5.8 (Low)	none mapped	-	-	-
Hays Composite Tooling	Approved	Medium	85.7	84	45,091	5.5 (Low)	none mapped	-	-	-
Newton Tool Grinding	Approved	Low	73.9	90	29,630	5.3 (Low)	none mapped	-	-	-
Topeka Test Labs	Approved	High	87	94	15,819	4.5 (Low)	none mapped	-	-	-
Dodge City Freight Services	Approved	Low	82.6	93	17,204	4.3 (Low)	none mapped	-	-	-
Hutchinson Seals & Gaskets	Approved	Medium	88.2	88	26,797	3.5 (Low)	none mapped	-	-	-
Lawrence Wire & Harness	Approved	Low	95.7	95	17,273	1.8 (Low)	none mapped	-	-	-
Sunflower Packaging & Kitting	Approved	Low	100	92	22,335	0.3 (Low)	none mapped	-	-	-

Supplier	Status	Criticality	OTD %	Quality	Reject PPM	Risk index	Flow-down	Special proc. without current approval	Sub-tier NADCAP gaps	Counterfeit open
Sunflower NDT Services LLC	Approved	-	-	0	-	-	none mapped	-	-	-

Suppliers needing attention

Great Plains Titanium — status probation; 1 sub-tier special process without a current NADCAP accreditation: Wichita Heat Treating Inc (nondestructive testing, not accredited); 1 counterfeit case open: CF-MER-003.

Heartland Aluminum Corp — status probation; 1 sub-tier special process without a current NADCAP accreditation: Flint Chemical Processing (chemical processing, expired 2026-08-31).

Prairie Composites LLC — status probation.

Wichita Heat Treating Inc — status conditional.

Central NDT Services — status conditional.

Flint Chemical Processing — status conditional.

Kansas Precision Castings — status conditional.

Titan Bar & Billet — status probation.

Aero Fastener Brokers — status conditional; 2 counterfeit cases open: CF-MER-001, CF-MER-002.

Great Bend Machining — status conditional; 1 sub-tier special process without a current NADCAP accreditation: Plains Surface Technologies (coatings, not accredited).

Salina Sheet Metal Works — status conditional.

Emporia Anodizing Co — status probation.

OTD % is the share of the last 12 months' receipts delivered on time. Reject PPM is rejected over received units on the last 12 months' receipts (24 of 25 suppliers have receipts). The risk index is Aegis's proprietary assessment of each supplier's delivery, quality and findings, taking account of the criticality your supplier list records. The table is ordered by it. Not assessed from the records supplied: on-time-in-full (receipts record quantity received, not quantity ordered); corrective-action request cycle time (quality events record no closure date).

9. Internal Audit Program Evidence

Internal audits: 8

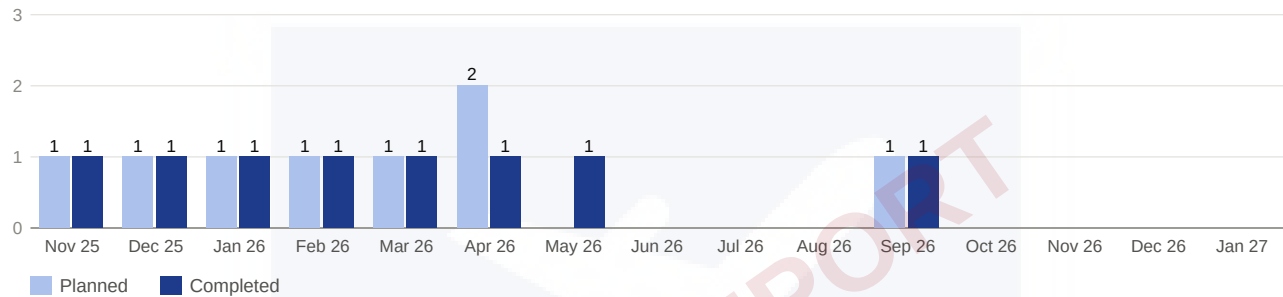
Completed: 8

In Progress: 0

Planned: 0

External, customer and registrar audits: 4 (listed in the schedule below; not part of the internal audit programme).

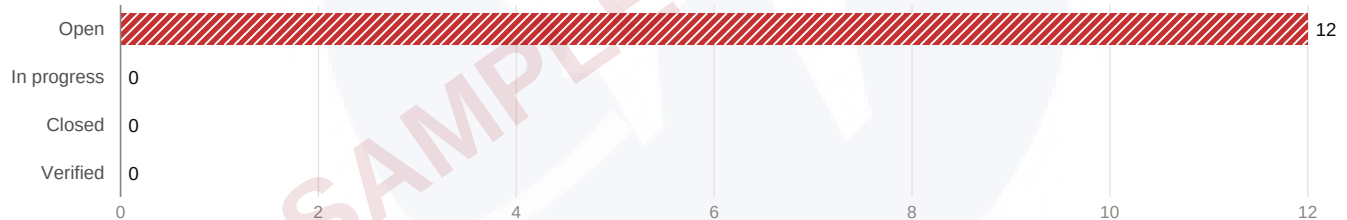
Internal audit programme: planned against completed



Internal audit programme: audits scheduled in each month and audits completed in it, twelve months back and three ahead.

8 internal audits scheduled and 8 completed from Nov 25 to Jan 27.

Findings by status



Audit findings by status, internal, external and customer.

12 audit findings: 12 open, 0 in progress, 0 closed, 0 verified.

Audit schedule

Audit	Name	Type	Status	Lead auditor	Scheduled	Findings
AUD-MER-01	Clause 4-6 context, leadership, risk	Internal	Completed	Registrar	2025-11-24	0
AUD-MER-02	Clause 7.1.5 measuring resources	Internal	Completed	P. Green	2025-12-20	0
AUD-MER-03	Clause 7.2 competence	Internal	Completed	K. Osei	2026-01-15	0
AUD-MER-04	Clause 8.4 external providers	Internal	Completed	K. Osei	2026-02-10	0
AUD-MER-05	Clause 8.5 production and traceability	Internal	Completed	K. Osei	2026-03-08	0
AUD-MER-06	Clause 8.7 / 10.2 nonconformity and CAPA	Internal	Completed	P. Green	2026-04-03	0
AUD-MER-07	Clause 9.1-9.3 performance evaluation	Internal	Completed	K. Osei	2026-04-29	7
AUD-MER-08	Boeing source inspection	External	Completed	P. Green	2026-05-25	2

Audit	Name	Type	Status	Lead auditor	Scheduled	Findings
AUD-MER-09	Nadcap heat treat	External	Completed	Registrar	2026-06-20	2
AUD-MER-10	CB surveillance (registrar)	Surveillance	Completed	Registrar	2026-07-16	0
AUD-MER-11	Airbus Level 3 escalation review	External	Completed	K. Osei	2026-08-11	0
AUD-MER-12	Clause 7.5 documented information	Internal	Completed	P. Green	2026-09-06	1

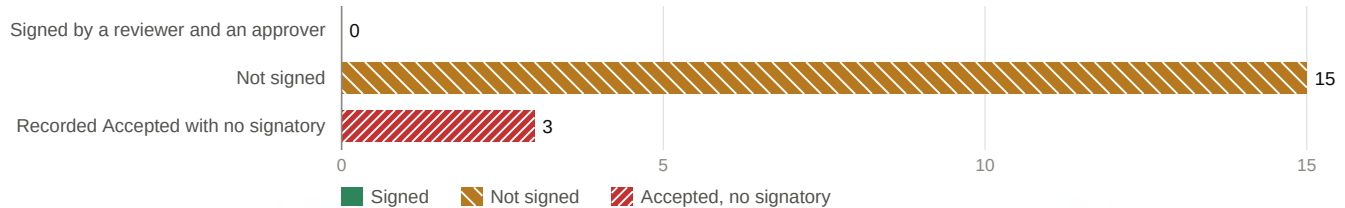
Findings Closure Rate: 0% (0/12)



10. First Article Inspection (AS9102)

Clause 8.5.1.3 — production process verification. Each report below shows its status and outcome, and whether it carries the two distinct signatures AS9102 Rev C fields 20 and 22 require.

Signature state



First article reports by signature state, from the FAI log. Accepted with no signatory is a records gap: the report was accepted but who reviewed and approved it is not on record.

18 first article reports: 0 signed, 15 not signed, 3 recorded Accepted with no signatory.

FAI	Part	Type	Customer	Status	Due	Findings / disposition	Signatures
FAI-MER-101	MER-3101 A320neo Nacelle Bracket Assy	Full	Airbus	Open	2026-10-02	Heat treat certs missing; 4 dimensional chars out of tolerance; Resubmit	Not signed
FAI-MER-102	MER-3102 737 Fuselage Frame Segment	Full	Boeing	Rejected	2026-09-24	Material CoC mismatch — wrong Ti alloy traced; Rejected — new lot required	Not signed
FAI-MER-103	MER-3103 Wing Rib Upper Panel	Full	Boeing	Open	2026-10-06	CNC program rev not matching drawing rev; In Review	Not signed
FAI-MER-104	MER-3104 Pylon Attach Fitting	Full	Airbus	Accepted	2026-09-14	1 minor balloon note error; Accepted	Accepted; no signatories recorded
FAI-MER-105	MER-3105 Floor Beam Support Clip	Partial	Boeing	Accepted	2026-09-01	No findings; Accepted	Accepted; no signatories recorded
FAI-MER-106	MER-3106 Composite Trailing Edge Panel	Full	Spirit AeroSystems	Open	2026-10-09	Autoclave cure cycle deviation; FPI rework not documented; Resubmit	Not signed
FAI-MER-107	MER-3107 Engine Mount Bracket	Delta	GE Aviation	Rejected	2026-09-26	Hardness below spec after heat treat; Rejected — process revalidation required	Not signed
FAI-MER-108	MER-3108 Cargo Door Hinge Fitting	Full	Airbus	Open	2026-10-12	Awaiting calibrated CMM data — CMM overdue; On Hold	Not signed
FAI-MER-109	MER-3109 Stringer Doubler	Delta	Boeing	Accepted	2026-09-06	1 minor cosmetic note; Accepted	Accepted; no signatories recorded
FAI-MER-110	MER-3110 Flap Track Fairing	Full	Spirit AeroSystems	Open	2026-10-17	Composite layup records incomplete; In Review	Not signed

FAI	Part	Type	Customer	Status	Due	Findings / disposition	Signatures
FAI-MER-111	MER-3111 Nacelle Thrust Link	Full	Airbus	Open	2026-09-23	Balloon drawing incomplete; In Review	Not signed
FAI-MER-112	MER-3112 Keel Beam Splice Plate	Full	Boeing	Open	2026-09-05	Balloon drawing incomplete; In Review	Not signed
FAI-MER-113	MER-3113 Access Panel Frame	Full	Spirit AeroSystems	Open	2026-09-25	Balloon drawing incomplete; In Review	Not signed
FAI-MER-114	MER-3114 Fan Cowl Latch Bracket	Full	GE Aviation	In review	2026-08-27	Balloon drawing incomplete; In Review	Not signed
FAI-MER-115	MER-3115 Seat Track Splice	Full	Boeing	In review	2026-08-06	Balloon drawing incomplete; In Review	Not signed
FAI-MER-116	MER-3116 Pressure Bulkhead Clip	Full	Airbus	Open	2026-09-16	Balloon drawing incomplete; In Review	Not signed
FAI-MER-117	MER-3117 Composite Fairing Skin	Full	Spirit AeroSystems	Open	2026-10-10	Balloon drawing incomplete; In Review	Not signed
FAI-MER-118	MER-3118 Titanium Hinge Lug	Full	Boeing	Open	2026-09-21	Balloon drawing incomplete; In Review	Not signed

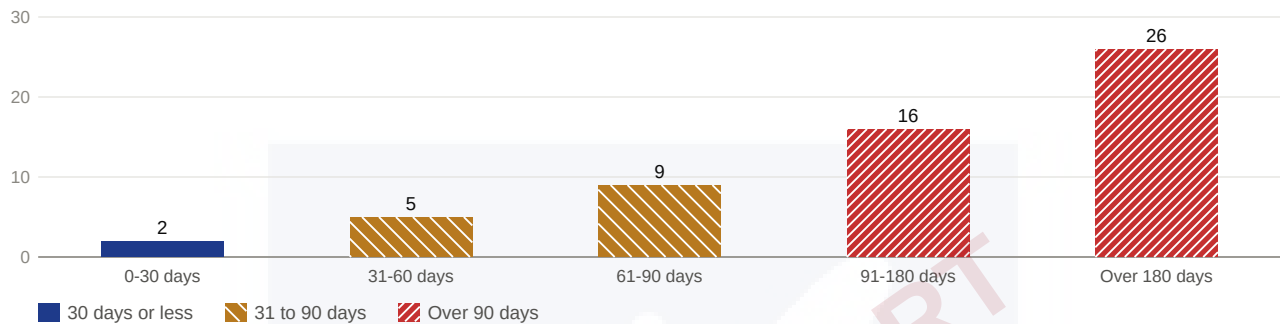
Open item: 15 first article reports are not yet signed by both a reviewer and an approver.
Records gap: 3 first article reports are recorded as accepted with no reviewer or approver on record.



11. Nonconforming Product, Containment and Material Review

Clause 8.7 — control of nonconforming outputs. Containment is the control that decides whether suspect product reached a customer, so it is reported per nonconformance rather than in summary.

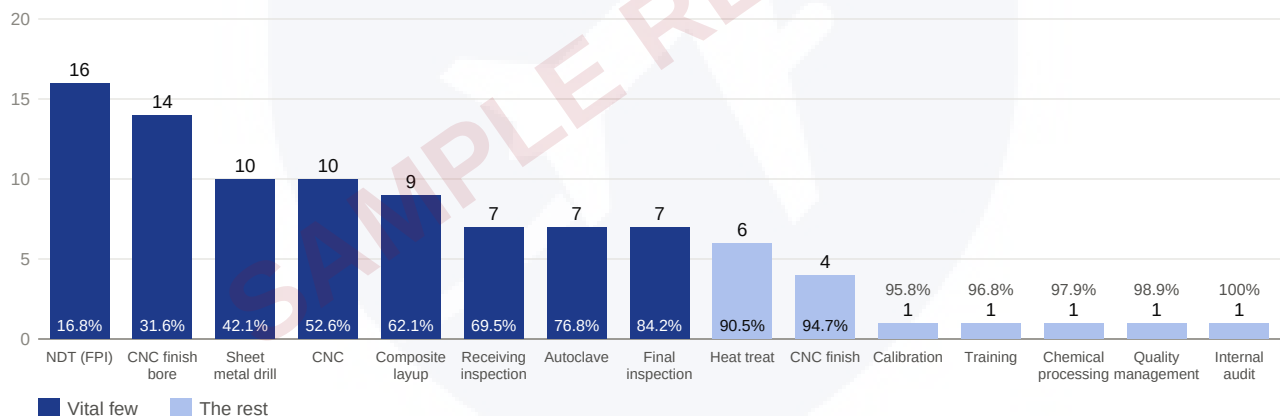
Open nonconformances by age



Open nonconformances by days since detected.

58 open nonconformances by age: 2 0-30 days, 5 31-60 days, 9 61-90 days, 16 91-180 days, 26 Over 180 days.

Where the nonconformances come from



95 nonconformances. Dark bars are the vital few (together 80%); the percentage in each bar is the cumulative share, largest first.

95 nonconformances: 8 of 15 account for 84.2%: NDT (FPI) (16), CNC finish bore (14), Sheet metal drill (10), CNC (10), Composite layup (9), Receiving inspection (7), Autoclave (7), Final inspection (7).

Containment status

NCR	Nonconformance	Contained	Zones outstanding	Deadline	Status
NCR-MER-001	Hardness below spec after heat treat — MER-3117	Yes	None	2026-06-29	Complete
NCR-MER-002	Hardness below spec after heat treat — MER-3106	Yes	None	2026-03-07	Complete
NCR-MER-003	Heat lot mismatch on CoC — MER-3114	Yes	None	2026-01-16	Complete
NCR-MER-004	Delamination — MER-3103	Yes	None	2026-09-23	Complete

NCR	Nonconformance	Contained	Zones outstanding	Deadline	Status
NCR-MER-005	Edge distance violation — MER-3109	Yes	None	2026-04-24	Complete
NCR-MER-006	Bore oversize — MER-3102	Yes	None	2025-12-03	Complete
NCR-MER-007	Surface finish Ra out of spec — MER-3114	Yes	None	2025-11-30	Complete
NCR-MER-008	Porosity at FPI — MER-3118	Yes	None	2025-12-16	Complete
NCR-MER-009	Delamination — MER-3102	Yes	None	2026-04-11	Complete
NCR-MER-010	Cure cycle deviation — MER-3117	Yes	None	2026-08-18	Complete
NCR-MER-011	Wrong revision program run — MER-3115	Yes	None	2026-09-14	Complete
NCR-MER-012	Porosity at FPI — MER-3103	Yes	None	2026-07-17	Complete
NCR-MER-013	Edge distance violation — MER-3106	Yes	None	2026-03-18	Complete
NCR-MER-014	Bore oversize — MER-3110	Yes	None	2026-08-18	Complete
NCR-MER-015	Edge distance violation — MER-3102	Yes	None	2026-07-06	Complete
NCR-MER-016	Heat lot mismatch on CoC — MER-3107	Yes	None	2026-04-10	Complete
NCR-MER-017	Porosity at FPI — MER-3115	Yes	None	2025-11-03	Complete
NCR-MER-018	Missing inspection stamp — MER-3105	Yes	None	2025-10-23	Complete
NCR-MER-019	Missing inspection stamp — MER-3115	Yes	None	2025-10-14	Complete
NCR-MER-020	Cure cycle deviation — MER-3102	Yes	None	2026-08-03	Complete
NCR-MER-021	Heat lot mismatch on CoC — MER-3107	Yes	None	2026-07-30	Complete
NCR-MER-022	Wrong revision program run — MER-3112	Yes	None	2026-06-30	Complete
NCR-MER-023	Hardness below spec after heat treat — MER-3104	Yes	None	2025-12-05	Complete
NCR-MER-024	Porosity at FPI — MER-3105	Yes	None	2026-06-05	Complete
NCR-MER-025	Edge distance violation — MER-3112	Yes	None	2026-05-18	Complete
NCR-MER-026	Edge distance violation — MER-3117	Yes	None	2026-05-21	Complete
NCR-MER-027	Cure cycle deviation — MER-3110	Yes	None	2026-03-20	Complete
NCR-MER-028	Porosity at FPI — MER-3109	Yes	None	2026-03-29	Complete
NCR-MER-029	Missing inspection stamp — MER-3103	Yes	None	2026-07-11	Complete
NCR-MER-030	Porosity at FPI — MER-3102	Yes	None	2026-01-06	Complete
NCR-MER-031	Wrong revision program run — MER-3110	Yes	None	2026-01-12	Complete
NCR-MER-032	Cure cycle deviation — MER-3112	Yes	None	2025-10-24	Complete
NCR-MER-033	Surface finish Ra out of spec — MER-3117	Yes	None	2026-02-18	Complete
NCR-MER-034	Cure cycle deviation — MER-3102	Yes	None	2025-12-24	Complete
NCR-MER-035	Delamination — MER-3109	Yes	None	2025-11-15	Complete
NCR-MER-036	Delamination — MER-3108	Yes	None	2026-01-04	Complete
NCR-MER-037	Delamination — MER-3110	Yes	None	2025-10-23	Complete
NCR-MER-038	Edge distance violation — MER-3116	Yes	None	2026-06-29	Complete
NCR-MER-039	Bore oversize — MER-3113	Yes	None	2026-06-06	Complete

NCR	Nonconformance	Contained	Zones outstanding	Deadline	Status
NCR-MER-040	Porosity at FPI — MER-3116	Yes	None	2026-04-06	Complete
NCR-MER-041	Bore oversize — MER-3107	Yes	None	2026-04-01	Complete
NCR-MER-042	Bore oversize — MER-3109	Yes	None	2025-12-06	Complete
NCR-MER-043	Delamination — MER-3103	Yes	None	2026-06-25	Complete
NCR-MER-044	Porosity at FPI — MER-3113	Yes	None	2026-05-17	Complete
NCR-MER-045	Porosity at FPI — MER-3118	Yes	None	2025-12-25	Complete
NCR-MER-046	Delamination — MER-3108	Yes	None	2026-06-08	Complete
NCR-MER-047	Surface finish Ra out of spec — MER-3103	Yes	None	2025-11-27	Complete
NCR-MER-048	Wrong revision program run — MER-3115	Yes	None	2026-03-28	Complete
NCR-MER-049	Bore oversize — MER-3113	Yes	None	2026-04-22	Complete
NCR-MER-050	Bore oversize — MER-3103	Yes	None	2025-11-12	Complete
NCR-MER-051	Wrong revision program run — MER-3101	Yes	None	2025-11-01	Complete
NCR-MER-052	Missing inspection stamp — MER-3110	Yes	None	2025-12-27	Complete
NCR-MER-053	Delamination — MER-3107	Yes	None	2025-12-03	Complete
NCR-MER-054	Surface finish Ra out of spec — MER-3111	Yes	None	2026-05-31	Complete
NCR-MER-055	Bore oversize — MER-3113	Yes	None	2025-12-25	Complete
NCR-MER-056	Porosity at FPI — MER-3103	Yes	None	2026-06-10	Complete
NCR-MER-057	Porosity at FPI — MER-3102	Yes	None	2026-03-27	Complete
NCR-MER-058	Delamination — MER-3101	Yes	None	2026-02-09	Complete
NCR-MER-059	Bore oversize — MER-3115	Yes	None	2026-08-19	Complete
NCR-MER-060	Bore oversize — MER-3114	Yes	None	2026-07-27	Complete
NCR-MER-061	Missing inspection stamp — MER-3101	Yes	None	2025-12-05	Complete
NCR-MER-062	Heat lot mismatch on CoC — MER-3113	Yes	None	2026-09-06	Complete
NCR-MER-063	Bore oversize — MER-3110	Yes	None	2026-05-19	Complete
NCR-MER-064	Bore oversize — MER-3110	Yes	None	2025-10-11	Complete
NCR-MER-065	Cure cycle deviation — MER-3112	Yes	None	2026-04-26	Complete
NCR-MER-066	Porosity at FPI — MER-3104	Yes	None	2025-11-03	Complete
NCR-MER-067	Wrong revision program run — MER-3114	Yes	None	2026-06-16	Complete
NCR-MER-068	Heat lot mismatch on CoC — MER-3108	Yes	None	2026-06-08	Complete
NCR-MER-069	Missing inspection stamp — MER-3117	Yes	None	2026-03-21	Complete
NCR-MER-070	Bore oversize — MER-3118	Yes	None	2026-01-11	Complete
NCR-MER-071	Edge distance violation — MER-3106	Yes	None	2026-08-03	Complete
NCR-MER-072	Wrong revision program run — MER-3113	Yes	None	2026-01-10	Complete
NCR-MER-073	Porosity at FPI — MER-3111	Yes	None	2026-07-28	Complete
NCR-MER-074	Porosity at FPI — MER-3116	Yes	None	2026-03-14	Complete

NCR	Nonconformance	Contained	Zones outstanding	Deadline	Status
NCR-MER-075	Hardness below spec after heat treat — MER-3115	Yes	None	2026-08-02	Complete
NCR-MER-076	Wrong revision program run — MER-3103	Yes	None	2026-07-14	Complete
NCR-MER-077	Edge distance violation — MER-3106	Yes	None	2026-05-17	Complete
NCR-MER-078	Wrong revision program run — MER-3114	Yes	None	2026-08-26	Complete
NCR-MER-079	Bore oversize — MER-3117	Yes	None	2026-01-09	Complete
NCR-MER-080	Porosity at FPI — MER-3105	Yes	None	2026-05-17	Complete
NCR-MER-201	Ti-6Al-4V forging inclusion — wrong heat lot traceability	Yes	None	2026-03-09	Complete
NCR-MER-202	Pyrometry gap — 6-month hole in heat treat furnace #3 records	Yes	None	2026-03-09	Complete
NCR-MER-203	A320neo nacelle bracket OTD failure — 4 consecutive months <60%	Yes	None	2026-03-09	Complete
NCR-MER-204	NDT FPI false-call rate 14% on aluminum wing skins	Yes	None	2026-03-09	Complete
NCR-MER-205	Composite layup cure cycle deviation — 3 autoclave batches out of spec	Yes	None	2026-03-09	Complete
NCR-MER-206	CNC program revision control — obsolete G-code run on 5-axis center	Yes	None	2026-03-09	Complete
NCR-MER-207	Calibration overdue — 32 assets including CMM and hardness testers	Yes	None	2026-03-09	Complete
NCR-MER-208	Expired operator certifications — 12 machinists and inspectors	Yes	None	2026-03-09	Complete
NCR-MER-209	Boeing rivet spacing nonconformance — fuselage panel lot 26-018	Yes	None	2026-03-09	Complete
NCR-MER-210	Chemical processing tank concentration drift — chromate conversion	Yes	None	2026-03-09	Complete
NCR-MER-211	Supplier raw material CoC discrepancies — 3 titanium lots	Yes	None	2026-03-09	Complete
NCR-MER-212	Prevent recurrence of quality manager vacancy gap	Yes	None	2026-03-09	Complete
NCR-MER-213	FOD control failure in sheet metal shop — debris in assembly	Yes	None	2026-03-09	Complete
NCR-MER-214	GE Aviation engine mount bracket hardness out of spec	Yes	None	2026-03-09	Complete
NCR-MER-215	Prevent audit finding recurrence — establish internal audit cadence	Yes	None	2026-03-09	Complete

Material review board dispositions

MRB	NCR	Part	Description	Disposition	Qty	Cost	Approver	Board date	Status
MRB-MER-301	NCR-MER-081	MER-3102	Ti-6Al-4V forging — inclusions found at ultrasonic inspection	Scrap	8	\$94,000	V. Singh (QD)	2026-08-25	Closed
MRB-MER-302	NCR-MER-082	MER-3101	Nacelle bracket — hardness 28 HRC vs 32-36 HRC spec after heat treat	Rework	12	\$38,000	G. Hoffman	2026-08-27	Closed
MRB-MER-303	NCR-MER-083	MER-3103	Wing rib — CNC profile 0.008" out of tolerance on contour	Rework	6	\$22,000	J. Reeves	2026-08-29	Closed
MRB-MER-304	NCR-MER-084	MER-3106	Composite panel — porosity >2% in ultrasonic C-scan	Scrap	4	\$67,000	S. Petrov	2026-08-31	Closed
MRB-MER-305	NCR-MER-085	MER-3105	Floor beam clip — rivet hole elongation from worn drill bushing	Rework	20	\$8,400	MRB Chair	2026-09-03	Closed
MRB-MER-306	NCR-MER-086	MER-3107	Engine mount bracket — Rockwell C hardness 3 points below minimum	Scrap	5	\$41,000	V. Singh (QD)	2026-09-05	Closed
MRB-MER-307	NCR-MER-087	MER-3101	Nacelle bracket lot 26-014 — chromate conversion coating blistering	Rework	15	\$18,500	P. Gutierrez	2026-09-07	Pending
MRB-MER-308	NCR-MER-088	MER-3108	Cargo door hinge — bore diameter 0.003" oversize	Use as is	3	\$1,200	Eng Auth (pending)	2026-09-10	Pending
MRB-MER-309	NCR-MER-089	MER-3102	Fuselage frame — FPI indication at fillet radius	Rework	7	\$29,000	M. Okonkwo	2026-09-12	Pending
MRB-MER-310	NCR-MER-090	MER-3110	Flap track fairing — fiber waviness exceeding 5° in ply 3	Scrap	3	\$52,000	S. Petrov	2026-09-14	Pending
MRB-MER-311	NCR-MER-091	MER-3104	Pylon fitting — anodize layer thickness below AMS 2468 minimum	Return to vendor	10	\$14,000	P. Gutierrez	2026-09-16	Pending
MRB-MER-312	NCR-MER-092	MER-3109	Stringer doubler — wrong aluminum alloy per supplier CoC	Return to vendor	18	\$27,000	K. Williams	2026-09-18	Pending
MRB-MER-313	NCR-MER-001	MER-3117	Hardness below spec after heat treat — MER-3117	Use as is	29	\$18,000	G. Hoffman	2026-07-01	Closed
MRB-MER-314	NCR-MER-003	MER-3114	Heat lot mismatch on CoC — MER-3114	Rework	3	\$38,000	G. Hoffman	2026-01-18	Closed
MRB-MER-315	NCR-MER-006	MER-3102	Bore oversize — MER-3102	Rework	7	\$11,000	G. Hoffman	2025-12-05	Closed

MRB	NCR	Part	Description	Disposition	Qty	Cost	Approver	Board date	Status
MRB-MER-316	NCR-MER-008	MER-3118	Porosity at FPI — MER-3118	Return to vendor	36	\$34,000	G. Hoffman	2025-12-18	Closed
MRB-MER-317	NCR-MER-009	MER-3102	Delamination — MER-3102	Return to vendor	7	\$21,000	G. Hoffman	2026-04-13	Closed
MRB-MER-318	NCR-MER-011	MER-3115	Wrong revision program run — MER-3115	Return to vendor	21	\$31,000	V. Singh (QD)	2026-09-16	Closed
MRB-MER-319	NCR-MER-013	MER-3106	Edge distance violation — MER-3106	Scrap	33	\$25,000	V. Singh (QD)	2026-03-20	Closed
MRB-MER-320	NCR-MER-018	MER-3105	Missing inspection stamp — MER-3105	Rework	1	\$18,000	G. Hoffman	2025-10-25	Closed
MRB-MER-321	NCR-MER-022	MER-3112	Wrong revision program run — MER-3112	Return to vendor	32	\$4,000	V. Singh (QD)	2026-07-02	Closed
MRB-MER-322	NCR-MER-023	MER-3104	Hardness below spec after heat treat — MER-3104	Rework	23	\$18,000	G. Hoffman	2025-12-07	Closed
MRB-MER-323	NCR-MER-026	MER-3117	Edge distance violation — MER-3117	Scrap	2	\$32,000	V. Singh (QD)	2026-05-23	Closed
MRB-MER-324	NCR-MER-029	MER-3103	Missing inspection stamp — MER-3103	Rework	7	\$54,000	V. Singh (QD)	2026-07-13	Closed
MRB-MER-325	NCR-MER-030	MER-3102	Porosity at FPI — MER-3102	Use as is	3	\$56,000	V. Singh (QD)	2026-01-08	Closed
MRB-MER-326	NCR-MER-032	MER-3112	Cure cycle deviation — MER-3112	Scrap	3	\$52,000	G. Hoffman	2025-10-26	Closed
MRB-MER-327	NCR-MER-035	MER-3109	Delamination — MER-3109	Use as is	17	\$15,000	V. Singh (QD)	2025-11-17	Closed
MRB-MER-328	NCR-MER-036	MER-3108	Delamination — MER-3108	Scrap	21	\$48,000	G. Hoffman	2026-01-06	Closed
MRB-MER-329	NCR-MER-037	MER-3110	Delamination — MER-3110	Rework	26	\$5,000	V. Singh (QD)	2025-10-25	Closed
MRB-MER-330	NCR-MER-039	MER-3113	Bore oversize — MER-3113	Rework	31	\$36,000	G. Hoffman	2026-06-08	Closed
MRB-MER-331	NCR-MER-043	MER-3103	Delamination — MER-3103	Rework	24	\$25,000	V. Singh (QD)	2026-06-27	Closed
MRB-MER-332	NCR-MER-046	MER-3108	Delamination — MER-3108	Return to vendor	13	\$15,000	V. Singh (QD)	2026-06-10	Closed
MRB-MER-333	NCR-MER-049	MER-3113	Bore oversize — MER-3113	Use as is	34	\$51,000	G. Hoffman	2026-04-24	Closed
MRB-MER-334	NCR-MER-050	MER-3103	Bore oversize — MER-3103	Rework	5	\$15,000	G. Hoffman	2025-11-14	Closed
MRB-MER-335	NCR-MER-051	MER-3101	Wrong revision program run — MER-3101	Return to vendor	28	\$60,000	G. Hoffman	2025-11-03	Closed
MRB-MER-336	NCR-MER-052	MER-3110	Missing inspection stamp — MER-3110	Scrap	20	\$4,000	G. Hoffman	2025-12-29	Closed

36 dispositions; recorded cost \$1,098,100.

5 dispositions are use-as-is or repair; either needs customer concession where the contract reserves it.



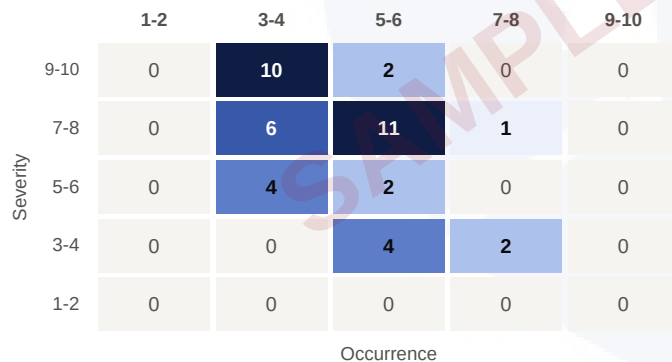
12. Advanced Product Quality Planning (AS9145)

Clause 8.1 — operational planning and control. Failure modes are ranked by Action Priority rather than risk priority number, because multiplying severity, occurrence and detection lets a flight-critical failure share a score with a cosmetic one.

Failure mode registers

FMEA	Part / Process	Type	Rev	Modes	High Priority	Status
PFMEA-MER-3101	MER-3101 A320neo Nacelle Bracket Assy machining and heat treat	PFMEA	B	7	4	In Review
PFMEA-MER-3102	MER-3102 737 Fuselage Frame Segment machining and heat treat	PFMEA	B	7	5	In Review
PFMEA-MER-3103	MER-3103 Wing Rib Upper Panel machining and heat treat	PFMEA	B	7	4	Draft
PFMEA-MER-3104	MER-3104 Pylon Attach Fitting machining and heat treat	PFMEA	B	7	4	Draft
PFMEA-MER-3105	MER-3105 Floor Beam Support Clip machining and heat treat	PFMEA	B	7	5	In Review
PFMEA-MER-3106	MER-3106 Composite Trailing Edge Panel machining and heat treat	PFMEA	B	7	5	In Review

Process failure modes by severity and occurrence



Count: 0 1 to 11, darker is more

42 process failure modes by severity and occurrence band, from the PFMEA registers. Severity 9-10 is the top row.

42 failure modes; 30 with severity 7 or above; 1 with severity and occurrence both 7 or above.

Control plans

Plan	Part	Stage	Rev	Characteristics	Status
CP-MER-3101	MER-3101 A320neo Nacelle Bracket Assy	Production	A	5	Draft
CP-MER-3103	MER-3103 Wing Rib Upper Panel	Production	A	5	Draft
CP-MER-3104	MER-3104 Pylon Attach Fitting	Production	A	5	Draft

Plan	Part	Stage	Rev	Characteristics	Status
CP-MER-3105	MER-3105 Floor Beam Support Clip	Production	A	5	Draft
CP-MER-3106	MER-3106 Composite Trailing Edge Panel	Production	A	5	Draft

Control plan flow-down (AS13100)

Finding: the failure modes below carry a High action priority and no control plan raised from their FMEA names them. A register and a control plan that disagree are two documents, not one control.

FMEA	Part / Process	High-priority modes	Not in a control plan
PFMEA-MER-3101	MER-3101 A320neo Nacelle Bracket Assy machining and heat treat	4	2
PFMEA-MER-3102	MER-3102 737 Fuselage Frame Segment machining and heat treat	5	5
PFMEA-MER-3103	MER-3103 Wing Rib Upper Panel machining and heat treat	4	2
PFMEA-MER-3104	MER-3104 Pylon Attach Fitting machining and heat treat	4	2
PFMEA-MER-3105	MER-3105 Floor Beam Support Clip machining and heat treat	5	2
PFMEA-MER-3106	MER-3106 Composite Trailing Edge Panel machining and heat treat	5	2

Production part approval

PPAP	Part	Customer	Level	Elements	Warrant	Status	Required
PPAP-MER-3101	MER-3101 A320neo Nacelle Bracket Assy	Airbus	3	None recorded	In Progress	Draft	2026-10-03
PPAP-MER-3102	MER-3102 737 Fuselage Frame Segment	Boeing	3	None recorded	Not Started	Draft	2026-09-11
PPAP-MER-3103	MER-3103 Wing Rib Upper Panel	Boeing	3	None recorded	In Progress	Draft	2026-09-13
PPAP-MER-3104	MER-3104 Pylon Attach Fitting	Airbus	3	None recorded	Not Started	Draft	2026-09-04
PPAP-MER-3105	MER-3105 Floor Beam Support Clip	Boeing	3	None recorded	In Progress	Submitted	2026-10-27
PPAP-MER-3106	MER-3106 Composite Trailing Edge Panel	Spirit AeroSystems	3	None recorded	In Progress	Draft	2026-11-18

Open item: 6 submissions have no signed Part Submission Warrant. Until the warrant is signed nobody has attested that the required elements are complete.

13. Process Capability and Measurement Systems

Clauses 9.1.1 and 7.1.5.2. Capability indices are derived from the recorded measurements, and control limits are computed from the process rather than the drawing tolerance. Cpk uses the within-subgroup spread and Ppk the overall spread, so a gap between them means the process is capable but not being held.

Capability studies

Study	Part	Characteristic	Spec	Cpk	Ppk	Chart	Gauge
SPC-3101-BORE	MER-3101	Bore diameter	12.687 - 12.713 mm	6.24	1.09	X-bar R	BG-14
SPC-3104-BORE	MER-3104	Bore diameter	18.98 - 19.02 mm	1.98	1.92	X-bar R	BG-6601
SPC-3101-FLANGE	MER-3101	Flange thickness	4.95 - 5.05 mm	2.05	1.96	X-bar R	MIC-7015
SPC-3108-HINGE	MER-3108	Hinge lug width	24.96 - 25.04 mm	1.66	1.16	X-bar R	CMM-4401
SPC-3107-HARD	MER-3107	Hardness HRC	32.0 - 36.0 HRC	0.81	0.81	X-bar R	HT-2205

1 characteristic shows Cpk materially above Ppk: the process is capable within a subgroup and drifting between them, which is a control problem rather than a capability problem.

Measurement system analysis

Study	Gauge	Gauge ID	Type	%GR&R	Repeat	Reprod	ndc	Expires
GRR-BG6601	Mitutoyo Bore Gauge	BG-6601	GR&R	5.0%	4.3%	2.5%	28	2027-05-07
GRR-CMM4401	Zeiss Contura CMM	CMM-4401	GR&R	23.9%	19.0%	14.5%	5	2027-05-07
GRR-HT2205	Wilson Rockwell	HT-2205	GR&R	59.0%	48.9%	33.2%	1	2027-05-07

Finding: the capability claims below rest on a measurement system that cannot carry them. A Cpk measured with a gauge consuming a third of the total variation is a statement about the gauge.

Characteristic	Part	Cpk	Gauge	Reason
Bore diameter	MER-3101	6.24	BG-14	gauge BG-14 has no Gage R&R study
Flange thickness	MER-3101	2.05	MIC-7015	gauge MIC-7015 has no Gage R&R study
Hardness HRC	MER-3107	0.81	HT-2205	gauge HT-2205 consumes 59.0% of total variation; this study measures the gauge

Measurements in doubt (7.1.5.2)

Finding: 3 calibration failures have no completed impact evaluation. Every measurement taken with the affected equipment since its last passing calibration remains in doubt, and product accepted on those measurements has not been dispositioned.

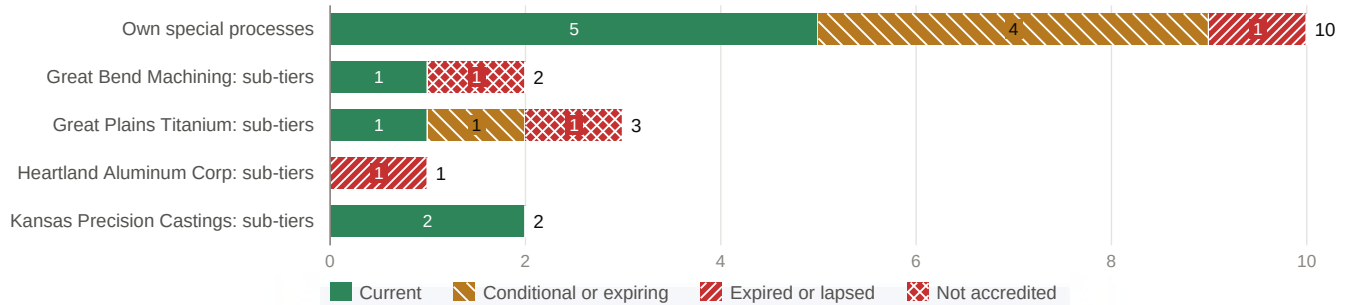
Asset	Equipment	Failed	Last good	Measurements in doubt	Evaluation
CAL-MER-121	Bore Gauge 12mm (cell 4)	2026-09-19	2026-01-17	Not yet assessed	Pending evaluation
CAL-MER-02	Wilson Rockwell Hardness Tester	2026-09-21	2026-02-03	Not yet assessed	Pending evaluation
CAL-MER-03	Fluke 1586A Super-DAQ Pyrometer	2026-09-21	2025-11-24	Not yet assessed	Pending evaluation

SAMPLE REPORT



14. Special Processes, Flow-down and Traceability

Special processes (8.5.1.2)



Special processes performed by each supplier's sub-tiers, and the organisation's own, by NADCAP or customer approval state, from the sub-tier map and the special-process register.

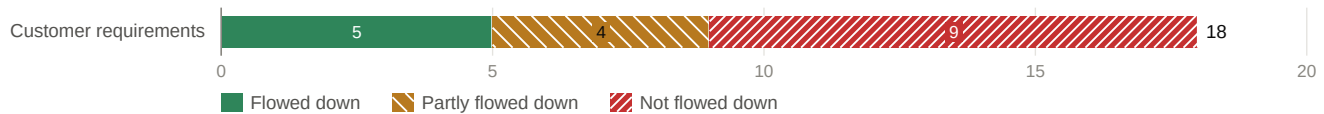
Own special processes: 5 current, 4 conditional or expiring, 1 expired or lapsed; Great Bend Machining: sub-tiers: 1 current, 1 not accredited; Great Plains Titanium: sub-tiers: 1 current, 1 conditional or expiring, 1 not accredited; Heartland Aluminum Corp: sub-tiers: 1 expired or lapsed; Kansas Precision Castings: sub-tiers: 2 current. 4 without a current approval.

Process	Specification	Certification body	NADCAP	Status	Expires	Audit findings
SP-MER-01 Heat Treat — Solution & Age (Ti-6Al-4V)	AMS 2801 / AMS 2750F	PRI	Conditional	Conditional	2026-11-29	3 Major: pyrometry records gap, furnace survey overdue, load thermocouple deviation
SP-MER-02 Heat Treat — Stress Relief (Aluminum)	AMS 2770 / AMS 2750F	PRI	Conditional	Conditional	2026-11-29	1 Major: pyrometry gap shared with SP-MER-01
SP-MER-03 NDT — Fluorescent Penetrant Inspection	ASTM E1417 / ASTM E2297	PRI	Current	Active	2027-03-06	2 Minor: operator qualification records, light meter calibration
SP-MER-04 NDT — Ultrasonic Testing	ASTM E2375 / AMS 2630	PRI	Current	Active	2027-02-13	1 Minor: reference standard traceability
SP-MER-05 NDT — Radiographic Inspection	ASTM E2104 / ASTM E1742	PRI	Expired	Expired	2026-09-14	2 Major: film density control, IQI selection documentation
SP-MER-06 Chemical Processing — Chromate Conversion	MIL-DTL-5541 / AMS 2473	PRI	Conditional	Conditional	2026-12-25	1 Major: tank concentration out-of-spec
SP-MER-07 Chemical Processing — Sulfuric Acid Anodize	MIL-A-8625 / AMS 2468	PRI	Current	Active	2027-04-01	1 Minor: rinse water conductivity
SP-MER-08 Shot Peening	AMS 2430 / AMS-S-13165	Customer Approved	N/A (customer approved)	Active	2027-04-17	No findings
SP-MER-09 Composite Autoclave Cure	Customer Spec + ASTM D5528	Customer Approved	N/A (customer approved)	Conditional	2026-12-14	1 Major: thermocouple array calibration overdue
SP-MER-10 Welding — GTAW Structural	AWS D17.1	Customer Approved	N/A (customer approved)	Active	2027-06-06	No findings

Open item: 1 special process is performed without a current approval.

Customer requirement flow-down (8.4.3)

18 customer requirements in scope: 5 flowed down to their supplier or in place in their process, 4 only partly, 9 not flowed down.



Accepted customer requirements by whether they reached the supplier or the process they apply to, from the flow-down register.

18 customer requirements: 5 flowed down, 4 only partly, 9 not flowed down.

Finding: the requirements below are not recorded as flowed to the supplier, or in place in the process, they apply to. A requirement the organisation accepted and did not pass on is a requirement nobody is meeting.

Requirement	Customer	Requirement text	Flowed to	Gap
FLD-MER-01	Boeing D6-82479	Counterfeit part prevention & traceability — AS9100D 8.4.3	Supplier: Great Plains Titanium	No counterfeit prevention clause in PO to supplier
FLD-MER-02	Boeing D6-82479	Material identification & heat lot traceability — AS9100D 8.5.2	Process: Receiving Inspection	Checklist does not include heat lot cross-reference step
FLD-MER-03	Airbus AIMS 09-00-002	Process control for special processes (heat treat) — AS9100D 8.5.1	Process: Heat Treat	Pyrometry procedure obsolete; no valid control document
FLD-MER-04	Airbus AIMS 04-04-011	Measurement system analysis for key characteristics — AS9100D 7.1.5.1	Process: Inspection	No MSA studies on file for nacelle bracket KC dimensions
FLD-MER-05	GE Aviation P11TF12	Product acceptance — hardness verification per AMS 6415 — AS9100D 8.6	Process: Heat Treat + Inspection	Hardness tester overdue on calibration (CAL-MER-02)
FLD-MER-11	Airbus AP2190	Key characteristic flowdown — AS9100D 8.4.3	Supplier: Titan Bar & Billet	Key characteristics not identified on POs to Titan Bar & Billet
FLD-MER-12	Airbus AP2190	Airbus FAI per AS9102 for all new parts — AS9100D 8.5.1	Process: FAI	FAI procedure does not require AS9102 FAI for every new Airbus part
FLD-MER-13	Spirit SQ-1000	Right of access for customer and regulator — AS9100D 8.4.3	Supplier: Great Bend Machining	Right-of-access clause missing from POs to Great Bend Machining
FLD-MER-15	GE S-1000	Special process sources must be GE-approved — AS9100D 8.4.3	Supplier: Emporia Anodizing Co	Emporia Anodizing Co not on GE's approved special-process source list

Only partly flowed down: started and not complete. These are not counted as flowed down.

Requirement	Customer	Requirement text	Partly flowed to	Gap
FLD-MER-09	Airbus AIMS 09-00-002	NADCAP heat treatment checklist compliance — NADCAP AC7102	Process: Heat Treat	Self-audit conducted but 3 checklist items non-compliant
FLD-MER-10	GE Aviation P11TF12	Operator competency requirements for special processes — AS9100D 7.2	Process: All special processes	12 expired certs not identified until post-vacancy review

Requirement	Customer	Requirement text	Partly flowed to	Gap
FLD-MER-16	Boeing D6-82479	Counterfeit prevention for distributors — AS9100D 8.1.4	Supplier: Aero Fastener Brokers	Not stated in the register
FLD-MER-18	Airbus AP2190	Gauge calibration traceable to national standards — AS9100D 7.1.5	Process: Calibration	Not stated in the register

Traceability and counterfeit prevention (8.5.2, 8.1.4)

40 certificates of conformity on record for this scope: 11 flagged, 29 verified.

Finding: 11 lots do not have a valid certificate of conformity. Material without traceable certification is suspect until proven otherwise (8.1.4).

Record	Supplier	Lot / heat	What was found
TRC-MER-01	Great Plains Titanium	LOT-26-009 / H-GP-7741	Missing — heat lot mismatch
TRC-MER-02	Great Plains Titanium	LOT-26-011 / H-GP-7742	Missing — CoC references wrong heat
TRC-MER-03	Great Plains Titanium	LOT-26-014 / H-GP-7745	Missing — no mill test report attached
TRC-MER-05	Prairie Composites LLC	LOT-26-018 / Batch-PC-441	Pending — shelf life verification
TRC-MER-15	Sunflower Packaging & Kitting	LOT-26-115 / H-4414	Missing — CoC not received
TRC-MER-19	Lawrence Wire & Harness	LOT-26-119 / H-2597	Missing — CoC not received
TRC-MER-22	Emporia Anodizing Co	LOT-26-122 / H-3780	Missing — CoC not received
TRC-MER-25	Prairie Composites LLC	LOT-26-125 / H-6314	Missing — CoC not received
TRC-MER-35	Aero Fastener Brokers	LOT-26-135 / H-4057	Missing — CoC not received
TRC-MER-36	Dodge City Freight Services	LOT-26-136 / H-2219	Missing — CoC not received
TRC-MER-38	Great Bend Machining	LOT-26-138 / H-6800	Missing — CoC not received

Counterfeit and suspect-part cases



Counterfeit and suspect-part cases by stage, from the traceability register. Open stages are hatched.

3 counterfeit or suspect-part cases, 3 open: investigation 1, quarantine 1, reporting 1.

Finding: 3 counterfeit part cases are recorded against this scope.

Case	Part	Supplier	Stage	What was found
CF-MER-001	MS21042-3	Aero Fastener Brokers	Quarantine	Head marking inconsistent with OEM
CF-MER-002	NAS1149F0363P	Aero Fastener Brokers	Investigation	CoC from unlisted distributor
CF-MER-003	AMS 4928 bar	Great Plains Titanium	Reporting	Heat lot genealogy cannot be established

15. Ownership, Actions and Management Review

Clauses 5.3, 9.3 and 10.2. An auditor asking who is accountable for a clause is asking for the RACI; asking what is being done about a finding is asking for the action register.

Assigned ownership (5.3)

1 RACI matrix defines accountability for the processes in scope.

Open actions (10.2)

Action	Owner	Due	Priority	Status
[ACT-MER-001] Establish daily recovery war room — 7 AM standup with all lane owners	Recovery Lead	2026-09-26	High	In Progress
[ACT-MER-005] Escalate titanium traceability issue to Boeing SQR — request containment guidance	R. Vasquez	2026-09-26	Critical	Not Started
[ACT-MER-004] Quarantine all product accepted with overdue calibration instruments	M. Okonkwo	2026-09-27	High	In Progress
[ACT-MER-002] Complete CAPA triage — rank all 24 open CAPAs by risk score and customer impact	V. Singh (QD)	2026-09-28	High	Not Started
[ACT-MER-013] Schedule Boeing and Airbus weekly recovery briefing cadence	T. Nakamura	2026-09-28	Medium	In Progress
[ACT-MER-003] Mobilize emergency calibration vendor — prioritize CMM, hardness tester, pyrometer	L. Tran	2026-09-29	High	Not Started
[ACT-MER-012] Build KPI scorecard baseline in Aegis AeroAssure® — establish truth dashboard	Recovery Lead	2026-09-29	Medium	Not Started
[ACT-MER-014] Escalate NADCAP radiographic inspection expiry — identify interim outsource capacity	M. Okonkwo	2026-09-29	Critical	Not Started
[ACT-MER-009] Escalate Airbus OTD recovery plan — request Level 3 review extension	T. Nakamura	2026-09-30	Critical	Not Started
[ACT-MER-007] Launch accelerated requalification program for 12 expired operator certifications	D. Kowalski	2026-10-02	High	Not Started
[ACT-MER-015] Map all open NCRs to root CAPAs and verify containment status	A. Chen	2026-10-02	Medium	Not Started
[ACT-MER-008] Prepare CB surveillance audit gap analysis — identify top 10 nonconformance risks	V. Singh (QD)	2026-10-04	High	Not Started
[ACT-MER-010] Establish pyrometry record restoration for Furnace #3 — 6-month gap closure	G. Hoffman	2026-10-06	High	Not Started

Quality objectives (9.1.1)

1 scorecard tracks quality objectives against target.

Management review (9.3)

Review	Date	Stage	Status
Quarterly business review	2026-10-08	Pre read	-

Management review records in the document register:

Document	Title	Type	Status
DOC-MER-034	RPT-220 Management Review Minutes	Record	Approved

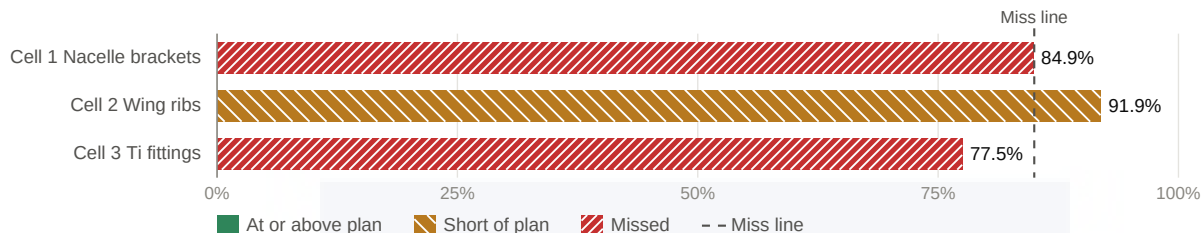
SAMPLE REPORT



16. Operations: the Floor, Material Review and the Supply Base

The floor, hour by hour

30 hour-by-hour boards: 1985 made against a plan of 2320 for the hours recorded (85.6% attainment); 126 hours missed and 49 escalations.



Actual as a share of plan for the hours recorded, every shift. Dashed: the miss line.

Attainment to plan: Cell 1 Nacelle brackets 84.9%, Cell 2 Wing ribs 91.9%, Cell 3 Ti fittings 77.5%.

Cell	09-21	09-22	09-23	09-24	09-25	Week	No reason written	Missed twice running
Cell 1 Nacelle brackets	84%	84%	91%	82%	83%	84.9%	6	10
Cell 2 Wing ribs	89%	93%	94%	90%	93%	91.9%	5	2
Cell 3 Ti fittings	79%	72%	83%	76%	77%	77.5%	14	12

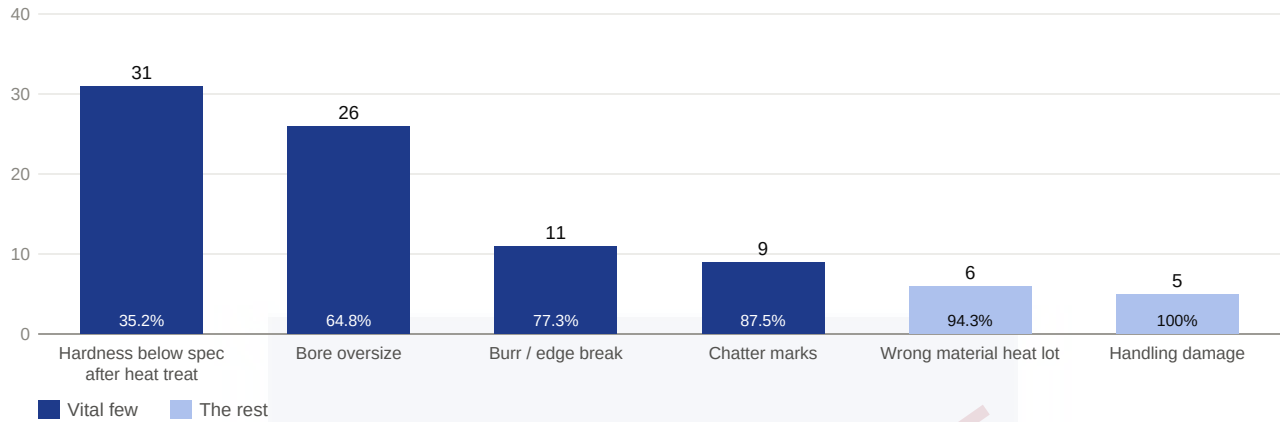
Every escalation, by hour, is on the Floor board page and in the Audit Defense Pack's 13_Operations workbook; 49 escalations this period.

Shift handover: issues still open

Cell	Logged	Shift	Type	Issue	Assigned to	Days open
Cell 3 Ti fittings	2026-09-22	1st	quality	Two MER-3102 forgings back from Wichita Heat Treating at 29 HRC; quarantined and MRB requested	Quality engineering	16
Cell 1 Nacelle brackets	2026-09-23	2nd	equipment	HMC-02 spindle alarm three times this shift; running at reduced feed	Maintenance	15
Cell 2 Wing ribs	2026-09-24	1st	material	Heartland plate lot 44718 short 6 sheets; Tuesday's ribs moved to Thursday	Purchasing	14
Cell 1 Nacelle brackets	2026-09-25	1st	quality	Bore gauge BG-14 read 0.004 mm high against the master; parts since 06:00 on hold for re-measure	Metrology	13

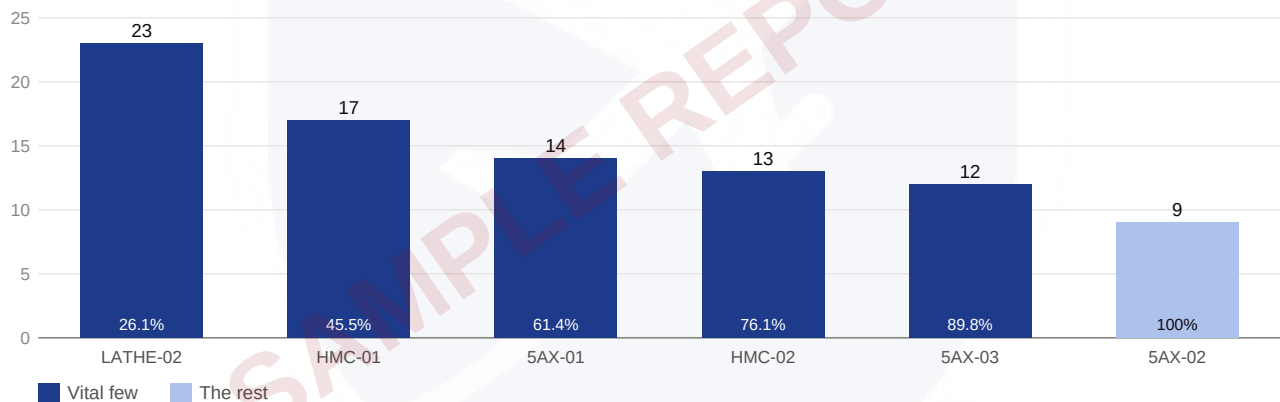
Where the scrap comes from

88 parts scrapped over 46 entries. 4 of 6 reasons make up 87.5%: Hardness below spec after heat treat (31), Bore oversize (26), Burr / edge break (11), Chatter marks (9).



88 parts scrapped, by reason. Dark bars are the vital few (together 80%); the percentage in each bar is the cumulative share, largest first.

88 parts scrapped, by reason: 4 of 6 account for 87.5%: Hardness below spec after heat treat (31), Bore oversize (26), Burr / edge break (11), Chatter marks (9).



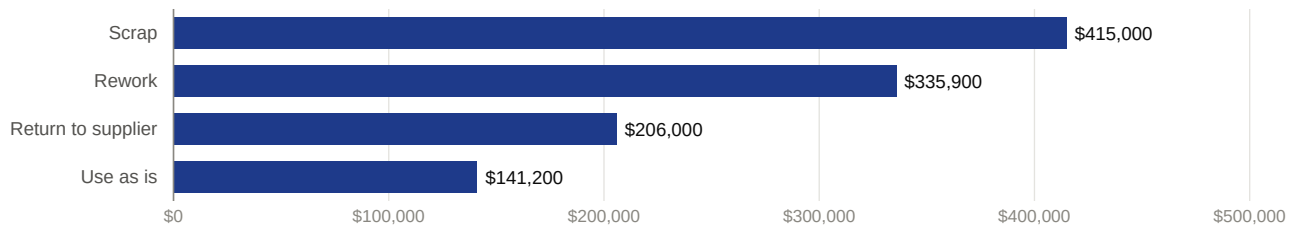
88 parts scrapped, by machine. Dark bars are the vital few (together 80%); the percentage in each bar is the cumulative share, largest first.

88 parts scrapped, by machine: 5 of 6 account for 89.8%: LATHE-02 (23), HMC-01 (17), 5AX-01 (14), HMC-02 (13), 5AX-03 (12).

Reason	Parts	Share	Cumulative	By machine
Hardness below spec after heat treat	31	35.2%	35.2%	HMC-02 8, HMC-01 7, 5AX-01 6, LATHE-02 6, 5AX-02 4
Bore oversize	26	29.5%	64.8%	HMC-01 7, LATHE-02 7, 5AX-03 6, 5AX-01 4, 5AX-02 2
Burr / edge break	11	12.5%	77.3%	LATHE-02 6, HMC-02 3, 5AX-03 2
Chatter marks	9	10.2%	87.5%	LATHE-02 4, 5AX-02 2, HMC-01 2, 5AX-03 1
Wrong material heat lot	6	6.8%	94.3%	5AX-03 3, 5AX-01 2, 5AX-02 1
Handling damage	5	5.7%	100%	5AX-01 2, HMC-02 2, HMC-01 1

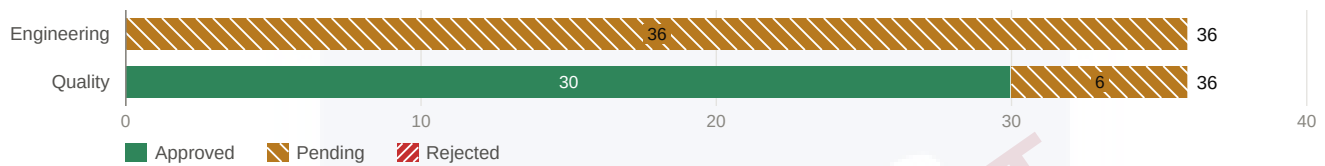
MRB: cost and approvals

36 dispositions with \$1,098,100 of cost impact recorded; 6 awaiting a board decision.



Cost impact recorded on the dispositions, by disposition.

MRB cost impact \$1,098,100: Scrap \$415,000, Rework \$335,900, Return to supplier \$206,000, Use as is \$141,200.



Dispositions by reviewer and decision. Customer counts only dispositions that need it.

MRB approvals: Engineering 0 approved, 36 pending, 0 rejected; Quality 30 approved, 6 pending, 0 rejected.

Disposition	Part	Type	Qty	Cost impact	Eng	Quality	Customer
NCR-MER-087	MER-3101	Rework	15	\$18,500	pending	pending	n/a
NCR-MER-088	MER-3108	Use as is	3	\$1,200	pending	pending	n/a
NCR-MER-089	MER-3102	Rework	7	\$29,000	pending	pending	n/a
NCR-MER-090	MER-3110	Scrap	3	\$52,000	pending	pending	n/a
NCR-MER-091	MER-3104	Return to supplier	10	\$14,000	pending	pending	n/a
NCR-MER-092	MER-3109	Return to supplier	18	\$27,000	pending	pending	n/a

Supplier onboarding



Supplier intakes by stage, from link sent to decision.

3 supplier intakes: 1 invited, 0 in progress, 1 submitted, 1 approved, 0 rejected.

Supplier	Stage	Red flags found	Decided by
Sunflower NDT Services	approved	None	V. Singh (QD)
Kansas Precision Plating	submitted	Special process 'chemical processing' offered without NADCAP accreditation for it; Special process 'coatings' offered without NADCAP accreditation for it; 1 root-cause analyses not closed: CAR-221; MER-3106: 40 good + 4 scrapped exceeds the 40 issued; Stated on-time delivery 98% but the imported data shows 50%	
Salina Anodize & Plate	invited	None	

Sub-tier special processes (NADCAP)

The accreditation of every sub-tier special process is charted in Section 14, beside the organisation's own.

Supplier	Sub-tier	Process	Status	Certificate	Expires
Great Bend Machining	Plains Surface Technologies	coatings	Not accredited	—	—
Great Plains Titanium	Central NDT Services	nondestructive testing	Expiring	NAD-NDT-11042	2026-10-15
Great Plains Titanium	Wichita Heat Treating Inc	nondestructive testing	Not accredited	—	—
Heartland Aluminum Corp	Flint Chemical Processing	chemical processing	Expired	NAD-CP-30911	2026-08-31

17. Data Completeness

Complete. Every register supplied for this report was read in full; no section is blank because records could not be read.

Every figure in this pack is drawn from the registers and records your organisation supplied, as they stood when the pack was generated. A section with nothing to report says so in words.

18. Appendices

A. Controlled Document List

Doc ID	Title	Type	Revision	Status	Owner
DOC-MER-001	QMS-001 Quality Management System Manual	Procedure	Rev 8	Approved	V. Singh (QD)
DOC-MER-002	HT-100 Heat Treat Operation — Titanium Alloys	Work instruction	Rev 4	Under review	G. Hoffman
DOC-MER-003	PYR-200 Pyrometry Control per AMS 2750F	Procedure	Rev 3	Obsolete	G. Hoffman
DOC-MER-004	RI-300 Receiving Inspection — Raw Material	Work instruction	Rev 6	Under review	K. Williams
DOC-MER-005	NDT-400 FPI Process Instruction per ASTM E1417	Work instruction	Rev 5	Approved	M. Okonkwo
DOC-MER-006	CAL-500 Calibration Control Procedure	Procedure	Rev 7	Under review	L. Tran
DOC-MER-007	MRB-600 Material Review Board Charter & Procedure	Procedure	Rev 3	Approved	V. Singh (QD)
DOC-MER-008	TRN-700 Training Qualification & Records Management	Procedure	Rev 2	Obsolete	D. Kowalski
DOC-MER-009	CNC-800 CNC Program Control & Verification	Work instruction	Rev 4	Under review	J. Reeves
DOC-MER-010	FAI-900 First Article Inspection Procedure per AS9102	Procedure	Rev 5	Approved	A. Chen
DOC-MER-011	FRM-101 Nonconformance Report Form	Form	Rev 6	Approved	V. Singh (QD)
DOC-MER-012	RPT-201 Monthly Quality Performance Report	Record	Rev 3	Under review	Recovery Lead
DOC-MER-013	QP-4.1 Context and Interested Parties	Procedure	Rev 4	Approved	G. Hoffman
DOC-MER-014	QP-5.3 Roles and Authorities	Procedure	Rev 9	Approved	V. Singh (QD)
DOC-MER-015	QP-6.1 Risk Management	Procedure	Rev 4	Approved	G. Hoffman
DOC-MER-016	QP-7.1.5 Monitoring and Measuring Resources	Procedure	Rev 2	Approved	D. Reyes
DOC-MER-017	QP-7.2 Competence	Procedure	Rev 7	Approved	S. Brooks
DOC-MER-018	QP-7.5 Control of Documented Information	Procedure	Rev 2	Approved	K. Osei
DOC-MER-019	QP-8.1.4 Counterfeit Parts Prevention	Procedure	Rev 4	Approved	K. Osei
DOC-MER-020	QP-8.4 Control of External Providers	Procedure	Rev 4	Under review	T. Nakamura
DOC-MER-021	QP-8.5.2 Identification and Traceability	Procedure	Rev 6	Approved	D. Reyes
DOC-MER-022	QP-8.5.6 Control of Changes	Procedure	Rev 4	Approved	M. Alvarez
DOC-MER-023	QP-8.7 Control of Nonconforming Output	Procedure	Rev 5	Approved	D. Reyes
DOC-MER-024	QP-9.1.2 Customer Satisfaction	Procedure	Rev 5	Approved	S. Brooks
DOC-MER-025	QP-9.2 Internal Audit	Procedure	Rev 2	Approved	G. Hoffman
DOC-MER-026	QP-9.3 Management Review	Procedure	Rev 3	Approved	G. Hoffman
DOC-MER-027	QP-10.2 Nonconformity and Corrective Action	Procedure	Rev 2	Approved	D. Reyes
DOC-MER-028	WI-CNC-810 Tool Life Management	Work instruction	Rev 2	Under review	L. Tran
DOC-MER-029	WI-SM-120 Drill and Rivet	Work instruction	Rev 9	Approved	T. Nakamura
DOC-MER-030	WI-COMP-330 Layup and Bagging	Work instruction	Rev 8	Approved	G. Hoffman

Doc ID	Title	Type	Revision	Status	Owner
DOC-MER-031	FRM-201 Supplier Scorecard	Form	Rev 2	Approved	M. Alvarez
DOC-MER-032	FRM-305 Calibration Recall Notice	Form	Rev 4	Approved	G. Hoffman
DOC-MER-033	FRM-410 FAI Report Forms 1-3	Form	Rev 4	Approved	L. Tran
DOC-MER-034	RPT-220 Management Review Minutes	Record	Rev 2	Approved	T. Nakamura
DOC-MER-035	WI-NDT-420 UT Inspection	Work instruction	Rev 3	Approved	G. Hoffman
DOC-MER-036	WI-HT-110 Furnace Load Thermocouples	Work instruction	Rev 5	Approved	R. Vasquez
DOC-MER-037	QP-8.3 Design Control (N/A statement)	Procedure	Rev 7	Approved	R. Vasquez
DOC-MER-038	QP-8.5.1.3 Production Process Verification	Procedure	Rev 3	Approved	D. Reyes
DOC-MER-039	QP-7.1.4 Work Environment and FOD	Procedure	Rev 6	Approved	P. Green
DOC-MER-040	FRM-150 Deviation Request	Form	Rev 2	Approved	L. Tran

B. Acronyms & Definitions

Acronym	Definition
AS9100	Quality Management Systems - Requirements for Aviation, Space and Defense Organizations
CAPA	Corrective and Preventive Action
FAI	First Article Inspection
FMEA	Failure Mode and Effects Analysis
GR&R	Gauge Repeatability and Reproducibility
IATF	International Automotive Task Force
ISO	International Organization for Standardization
KPI	Key Performance Indicator
MRB	Material Review Board
MSA	Measurement System Analysis
NCR	Nonconformance Report
NDT	Non-Destructive Testing
OJT	On-the-Job Training
OTD	On-Time Delivery
QBR	Quarterly Business Review
QMS	Quality Management System
RACI	Responsible, Accountable, Consulted, Informed
RCA	Root Cause Analysis
SOP	Standard Operating Procedure

C. Pack Generation

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D. Your Audit Checklist

106 questions an AS9100D auditor asks about the 15 clauses where your own records show a gap, critical first. Walk each clause with its process owner before the audit: tick a question only when the evidence is in hand, and note where it is kept. The full set of 252 questions is in the ZIP (11_Self_Audit/AS9100D_Self_Audit_Questions.xlsx), with these marked.

Clause 8.4.2: Type and extent of control

What your records showed

What was found	Records	Which
Sub-tier special process without NADCAP accreditation	3	Heartland Aluminum Corp, Great Bend Machining, Great Plains Titanium
Sub-tier NADCAP accreditation expiring	1	Great Plains Titanium
FIND-MER-007 Supplier CoC discrepancies — 3 titanium lots from Great Plains Titanium with formatting errors	-	

	Question	Priority	Evidence location / initials
☐	QB-8.4.2-002 Are customer and regulatory flow-down requirements included in purchase orders to external providers?	Critical	
☐	QB-8.4.2-001 Are purchasing requirements adequately defined and communicated to suppliers?	High	
☐	QB-8.4.2-003 Do purchase orders contain complete and unambiguous technical requirements including revision levels?	High	

	Have ready		
☐	Purchase orders with flow-down quality clauses	☐	Quality clauses in supplier agreements
☐	Flow-down requirements checklist or matrix	☐	Evidence of right-of-access provisions
☐	Sub-tier special process requirements records	☐	Purchase order samples with technical requirements
☐	Right-of-access clause evidence	☐	Drawing/specification revision references in POs
☐	Purchase order samples with requirement flowdown	☐	Special process requirements in POs
☐	Supplier quality requirements document	☐	PO review and approval records

Red flags an auditor looks for	
No flow-down quality clauses in purchase orders	No flowdown of critical requirements to sub-tier suppliers
Sub-tier suppliers unaware of prime customer requirements	Purchase orders lack drawing/specification references
Purchase orders contain only part number and quantity	Supplier produced to wrong revision due to PO error

Clause 8.1.4: Prevention of counterfeit parts (AS9100 specific)

What your records showed

What was found	Records	Which
Counterfeit part case open	3	CF-MER-001, CF-MER-002, CF-MER-003

	Question	Priority	Evidence location / initials
☐	QB-8.1.4-001 Are counterfeit part prevention controls implemented per AS6174/AS6081?	Critical	
☐	QB-8.1.4-002 Is counterfeit part prevention integrated into the procurement and receiving inspection processes?	Critical	
☐	QB-8.1.4-003 Are obsolete or surplus electronic components managed to prevent counterfeit parts from entering the supply chain?	Critical	
☐	QB-8.1.4-004 Is there a process for reporting suspect unapproved parts (SUPs) to the FAA or equivalent authority?	Critical	

	Have ready		
☐	Counterfeit parts prevention procedure	☐	Electronic component sourcing procedure
☐	Approved supplier list with source verification	☐	Authorized distributor list for electronic components
☐	Incoming inspection records for counterfeit detection	☐	Enhanced inspection procedure for non-standard sources
☐	Training records on counterfeit awareness	☐	X-ray or decapsulation testing records for suspect components
☐	Quarantine and disposition records for suspect parts	☐	SUP reporting procedure
☐	GIDEP or equivalent reporting records	☐	SUP investigation records
☐	Approved supplier list specifying authorized distributors	☐	FAA/authority reporting records
☐	Authentication testing procedures and records	☐	GIDEP reporting records for confirmed issues
☐	GIDEP alert monitoring records	☐	Quarantine and disposition records for SUPs
☐	Quarantine records for suspected counterfeit parts		

Red flags an auditor looks for	
Parts purchased from brokers without source verification	GIDEP membership lapsed or alerts ignored
No counterfeit prevention procedure exists	Electronic components purchased from brokers without testing
Suspect parts found but not reported	No counterfeit detection testing capability for electronics
Parts purchased from unapproved sources without additional testing	No SUP reporting procedure despite regulatory requirement
No counterfeit part prevention program exists	SUPs found but not reported to authorities

Clause 8.5.1.2: Validation and control of special processes

What your records showed

What was found	Records	Which
Special process without current accreditation	1	NDT — Radiographic Inspection

	Question	Priority	Evidence location / initials
☐	QB-NADCAP-001 Are special processes (welding, heat treat, NDT, surface treatment) validated per NADCAP requirements?	Critical	
☐	QB-NADCAP-002 Are NDT (Non-Destructive Testing) operations controlled per NAS-410/EN4179 and customer requirements?	Critical	
☐	QB-NADCAP-003 Are heat treatment processes controlled with continuous monitoring and recording of parameters?	Critical	
☐	QB-NADCAP-004 Are NDT procedures qualified per applicable specifications and are procedure qualification records maintained?	Critical	
☐	QB-NADCAP-005 Is the chemical process control (plating, anodizing, passivation) maintained per NADCAP Chemical Processing requirements?	Critical	
☐	QB-NADCAP-006 Are welding procedures qualified per applicable codes (AWS D17.1, AMS standards) and are welder qualifications current?	Critical	
☐	QB-NADCAP-007 Is composite material processing controlled per NADCAP requirements including layup, cure, and NDT inspection?	Critical	
☐	QB-NADCAP-008 Are shot peening and surface enhancement processes controlled per AMS 2430/2432 and NADCAP requirements?	Critical	
☐	QB-NADCAP-010 Are special process suppliers NADCAP accredited or equivalently qualified for the specific processes they perform?	Critical	
☐	QB-NADCAP-011 Are NDT operator eye examinations current and do they meet the requirements of NAS 410/EN 4179?	Critical	
☐	QB-NADCAP-012 Are heat treatment furnace instrumentation systems compliant with AMS 2750 (pyrometry) requirements?	Critical	
☐	QB-NADCAP-013 Are NDT sensitivity levels (penetrant, magnetic particle, ultrasonic) verified before each inspection session?	Critical	
☐	QB-NADCAP-015 Are non-conventional machining processes (EDM, ECM, laser, waterjet) controlled per customer and specification requirements?	Critical	
☐	QB-NADCAP-009 Are NADCAP audit findings addressed with effective corrective actions and are root causes systemic?	High	
☐	QB-NADCAP-014 Are coating (paint, primer, sealant) application processes controlled per customer and specification requirements?	High	

Have ready	
☐ NADCAP accreditation certificate (if applicable)	☐ Almen strip intensity verification records
☐ Process validation records	☐ Coverage verification records
☐ Process parameter monitoring records	☐ Media condition monitoring records
☐ Operator qualification records	☐ Supplier NADCAP accreditation certificates with scope

Have ready			
☐	Equipment calibration/validation records	☐	Equivalent qualification records for non-NADCAP suppliers
☐	NDT personnel certification records (NAS-410/EN4179)	☐	Customer approval records for special process suppliers
☐	Level III oversight records	☐	Supplier audit records for special processes
☐	NDT procedure approvals	☐	Annual eye examination records for all NDT personnel
☐	Equipment calibration records	☐	Color vision test results
☐	Sensitivity verification/system performance records	☐	Near-vision acuity test results (Jaeger J1 or equivalent)
☐	AMS2750 pyrometry records	☐	Corrective lens documentation on certification records
☐	Temperature uniformity survey (TUS) reports	☐	AMS 2750 compliance checklist for each furnace
☐	System accuracy test (SAT) records	☐	Instrument type classification records
☐	Thermocouple calibration records	☐	SAT records with pass/fail results
☐	Furnace chart records showing process parameters	☐	TUS records with uniformity data
☐	NDT procedure qualification records	☐	Thermocouple expenditure records (load vs. control)
☐	Reference standard/test piece records and calibration	☐	System performance check records
☐	Procedure qualification review and approval records	☐	Reference standard certification and verification
☐	Customer or prime approval of NDT procedures	☐	Sensitivity verification logs
☐	Solution analysis records and control charts	☐	Out-of-sensitivity response records
☐	Coupon test results per specification	☐	Non-conventional machining process specifications
☐	Chemical process specification compliance matrix	☐	Etch inspection records for EDM/ECM surfaces
☐	Waste treatment and environmental compliance records	☐	Metallographic examination records where required
☐	WPS documentation with supporting PQR	☐	NADCAP audit reports and findings
☐	Welder qualification records with expiration dates	☐	Corrective action responses to NADCAP findings
☐	Weld inspection procedure and inspector qualifications	☐	NADCAP accreditation certificates
☐	Weld parameter monitoring records	☐	Self-audit results (pre-audit preparation)
☐	Weld repair procedure and records	☐	Coating material traceability records
☐	Cure cycle monitoring records (autoclave or oven)	☐	Shelf-life monitoring for coatings and sealants
☐	Material traceability records including shelf-life and out-time	☐	Application parameter monitoring records
☐	Ply layup verification records	☐	Film thickness measurement records
☐	NDT inspection of composite parts (ultrasonic, thermography)	☐	Operator qualification records for coating application
☐	Material qualification test results		

Red flags an auditor looks for	
Special process performed without validation	Coverage requirements not met or not verified
NADCAP accreditation expired	Special processes outsourced to non-accredited, non-qualified suppliers
Operators performing special processes without current qualification	Supplier NADCAP accreditation expired without organization's knowledge
NDT performed by uncertified personnel	NDT operators performing inspections with expired eye exams

Red flags an auditor looks for	
No Level III available for procedure approval	Color blind NDT operator performing liquid penetrant inspection
TUS failed but furnace continued in operation	Furnace operating with expired TUS or SAT
No pyrometry compliance records	AMS 2750 compliance not maintained
NDT performed with unqualified procedures	Inspections performed without sensitivity verification
Reference standards missing or unverified	Reference standards uncertified or damaged
Solution analysis overdue or control limits exceeded	No etch inspection after EDM on flight-critical parts
No coupon testing for thickness, adhesion, or corrosion resistance	Excessive recast layer without detection or disposition
Welding performed without qualified WPS	NADCAP accreditation lapsed or at risk
Welders performing work outside their qualification range	Same findings recurring at multiple NADCAP audits
Prepreg material used beyond allowable out-time or shelf-life	Expired coating materials used in production
Cure cycle records show deviations with no investigation	Environmental conditions out of specification during coating application
Shot peening performed without intensity verification	

Clause 8.5.1.3: Production process verification

What your records showed

What was found	Records	Which
Part submission warrant unsigned	6	MER-3102 (PPAP-MER-3102), MER-3103 (PPAP-MER-3103), MER-3104 (PPAP-MER-3104), MER-3101 (PPAP-MER-3101), MER-3105 (PPAP-MER-3105), MER-3106 (PPAP-MER-3106)
First article report not signed	15	MER-3117 (FAI-MER-117), MER-3118 (FAI-MER-118), MER-3101 (FAI-MER-101), MER-3115 (FAI-MER-115), MER-3112 (FAI-MER-112), MER-3106 (FAI-MER-106), MER-3102 (FAI-MER-102), MER-3107 (FAI-MER-107), MER-3103 (FAI-MER-103), MER-3116 (FAI-MER-116), MER-3114 (FAI-MER-114), MER-3110 (FAI-MER-110), MER-3108 (FAI-MER-108), MER-3111 (FAI-MER-111), MER-3113 (FAI-MER-113)
Accepted first article report with no signatures on record	3	MER-3104 (FAI-MER-104), MER-3109 (FAI-MER-109), MER-3105 (FAI-MER-105)

	Question	Priority	Evidence location / initials
☐	QB-AS9102-001 Is First Article Inspection (FAI) performed per AS9102 for new or changed parts?	Critical	
☐	QB-AS9102-004 Are FAI triggers clearly defined and consistently applied (new part, design change, process change, supplier change, production break)?	Critical	
☐	QB-AS9102-005 Does the FAI process include verification of all design characteristics including functional, dimensional, and material requirements?	Critical	
☐	QB-AS9102-015 Are FAI measurement results compared against design tolerances with clear pass/fail determination for each characteristic?	Critical	
☐	QB-AS9102-002 Are FAI records complete with all required forms and data?	High	
☐	QB-AS9102-006 Are FAI non-conformances dispositioned and is the FAI re-performed or partially re-performed after corrective action?	High	

	Question	Priority	Evidence location / initials
☐	QB-AS9102-007 Are FAI records maintained with traceability to the specific part, tooling, equipment, and operators used?	High	
☐	QB-AS9102-008 Is the organization managing FAI requirements flow-down to sub-tier suppliers and verifying their FAI completion?	High	
☐	QB-AS9102-010 Is the FAI Form 1 (Part Number Accountability) complete with all required information including design authority and part revision?	High	
☐	QB-AS9102-011 Is FAI Form 2 (Raw Material, Special Processes, and Functional Testing) properly completed with actual test results?	High	
☐	QB-AS9102-012 Are FAI results reviewed and approved by authorized personnel with appropriate technical competence?	High	
☐	QB-AS9102-013 Is the FAI process integrated with the organization's change management process to ensure FAI is triggered by all applicable changes?	High	
☐	QB-AS9102-003 Is partial FAI properly used and documented when only some characteristics are affected by a change?	Medium	
☐	QB-AS9102-009 Is there a process for managing FAI waivers or deviations from the standard FAI process?	Medium	
☐	QB-AS9102-014 Is FAI status tracked and reported including open FAIs, overdue FAIs, and FAI pass/fail rates?	Medium	

Have ready	
☐ AS9102 FAI procedure	☐ Sub-tier supplier FAI reports received
☐ Completed FAI forms (Form 1: Part Number Accountability, Form 2: Product Accountability, Form 3: Characteristic Accountability)	☐ Review and acceptance records for sub-tier FAIs
☐ FAI trigger matrix showing when FAI is required	☐ Sub-tier FAI tracking log
☐ Partial FAI records (when applicable)	☐ FAI Form 1 with all fields completed
☐ Sub-tier FAI records or delegation evidence	☐ Cross-reference to sub-assembly FAIs
☐ FAI trigger criteria procedure	☐ Manufacturing process reference in Form 1
☐ FAI tracking log showing trigger reasons	☐ Design authority and revision identification
☐ Production break monitoring records	☐ FAI Form 2 with material data populated
☐ Supplier change notification records triggering FAI	☐ Material certification cross-references
☐ Ballooned drawings with characteristic numbering	☐ Special process certification references
☐ FAI Form 3 with all characteristics measured	☐ Functional test results with actual values
☐ Material certification verification in FAI	☐ FAI approval authority designation
☐ Special process certification verification in FAI	☐ Approver competence/qualification records
☐ FAI Form 3 with design values and actual measurements	☐ Multi-level approval records (if required)
☐ Pass/fail determination for each characteristic	☐ FAI rejection and re-work records
☐ Near-limit analysis for characteristics close to tolerance	☐ Change management procedure referencing FAI requirements
☐ Measurement uncertainty consideration in pass/fail decisions	☐ ECN/ECO forms with FAI impact assessment field
☐ Completed Form 1 with all required fields	☐ FAI trigger evaluation records for recent changes

Have ready			
☐	Completed Form 2 with material and process details	☐	Supplier change notifications linked to FAI requirements
☐	Completed Form 3 with actual measured values for all characteristics	☐	Partial FAI records referencing original FAI
☐	Supporting raw data (CMM reports, test results)	☐	Change justification for partial FAI
☐	FAI non-conformance disposition records	☐	Approval records for partial FAI
☐	Corrective action records for FAI failures	☐	FAI waiver/deviation procedure
☐	Re-FAI or partial re-FAI records after corrective action	☐	FAI waiver records with justification
☐	Customer notification of FAI failures if required	☐	Customer approval for FAI deviations
☐	FAI records with part serial/lot number traceability	☐	Risk assessment supporting FAI waivers
☐	Measurement equipment identification in FAI records	☐	FAI tracking dashboard or log
☐	Operator identification in FAI documentation	☐	FAI aging report
☐	Tooling and fixture identification used during FAI production	☐	FAI pass/fail rate metrics
☐	Purchase orders with FAI flow-down requirements	☐	Management review inputs on FAI status
Red flags an auditor looks for			
No FAI process in place		Sub-tier suppliers not performing FAI when required	
FAI not re-performed after design changes		Sub-tier FAI reports never reviewed or accepted	
Missing characteristics on Form 3		Form 1 incomplete with missing required fields	
Production resumed after long break without FAI		Wrong design revision referenced in FAI	
No defined production break threshold		Form 2 incomplete or missing sections	
Significant number of characteristics missing from FAI		Material data not verified against actual certifications	
Material verification not part of FAI process		FAIs approved by unauthorized personnel	
FAI approved with characteristics outside tolerance		Complex FAIs approved without engineering review	
Actual measurements missing or incorrect on Form 3		Changes made without FAI evaluation	
Forms incomplete or missing required data fields		No integration between change management and FAI process	
No actual measurements recorded, only pass/fail		Partial FAI used without a valid original complete FAI	
FAI approved with out-of-tolerance characteristics and no concession		FAI waivers granted without customer knowledge or approval	
FAI failures without corrective action		Blanket FAI waivers without case-by-case evaluation	
FAI records lack traceability to the inspected part		Large number of overdue FAIs with no plan to complete	
Measurement equipment used during FAI found out of calibration		No FAI tracking or reporting system	

Clauses 7.1.5, 7.1.5.1 and 7.1.5.2: Measurement traceability

What your records showed

What was found	Records	Which
Measurements in doubt after a calibration failure	3	Wilson Rockwell Hardness Tester (CAL-MER-02), Fluke 1586A Super-DAQ Pyrometer (CAL-MER-03), Bore Gauge 12mm (cell 4) (CAL-MER-121)
Capability claim resting on an unqualified gauge	3	Hardness HRC on MER-3107 measured with HT-2205, Bore diameter on MER-3101 measured with BG-14, Flange thickness on MER-3101 measured with MIC-7015
Customer requirement not flowed down	1	FLD-MER-04 (Airbus) in Inspection
FIND-MER-004 32 calibration assets overdue — measurement system integrity compromised	-	

	Question	Priority	Evidence location / initials
☐	QB-7.1.5-001 Are monitoring and measuring resources identified, controlled, and calibrated?	Critical	
☐	QB-7.1.5-004 Is calibration traceable to national or international standards and are calibration intervals justified?	Critical	
☐	QB-7.1.5-006 Is there a process for handling calibration recall when equipment is found out of tolerance?	Critical	
☐	QB-7.1.5-002 Is measurement uncertainty considered for critical measurements?	High	
☐	QB-7.1.5-003 Are measurement systems analyzed for adequacy using Measurement System Analysis (MSA) techniques?	High	
☐	QB-7.1.5-005 Are measurement devices suitable for the measurement task in terms of accuracy, resolution, and range?	High	
☐	QB-7.1.5-007 Are customer-owned or customer-specified measurement devices controlled and calibrated per customer requirements?	High	

Have ready	
☐ Calibration schedule/register	☐ Gauge R&R study results
☐ Calibration certificates with traceability	☐ Uncertainty budget calculations for critical measurements
☐ Out-of-calibration investigation records	☐ Gage R&R study results for critical gages
☐ Equipment identification labels/stickers	☐ MSA plan identifying which instruments require studies
☐ Recall procedure for OOC equipment	☐ Action records for unacceptable MSA results
☐ Calibration certificates with traceability statements	☐ Measurement uncertainty budgets for critical measurements
☐ Calibration interval justification records	☐ Measurement device selection criteria
☐ Out-of-tolerance notification and impact assessment records	☐ Device capability verification records
☐ Calibration provider accreditation (ISO 17025)	☐ Resolution and accuracy specifications vs. requirements
☐ Out-of-tolerance recall procedure	☐ Customer-specified gage register
☐ Impact assessment records for out-of-tolerance findings	☐ Calibration records per customer requirements

Have ready			
☐	Product disposition records for affected measurements	☐	Customer-owned gage identification and tracking
☐	Customer notification records when product is affected	☐	Customer gage maintenance records
☐	Measurement system analysis records		
Red flags an auditor looks for			
Multiple instruments overdue for calibration		No awareness of measurement uncertainty requirements	
No process for handling out-of-calibration discoveries		No MSA program exists	
Calibration performed by unqualified personnel		Critical measurements made with gages showing >30% R&R	
Gages in use past calibration due date		Measurement devices with insufficient resolution for critical dimensions	
No impact assessment process for out-of-tolerance findings		No consideration of measurement capability in device selection	
Calibration provider lacks accreditation		Customer-specified calibration requirements not followed	
No recall process for out-of-tolerance findings		Customer-owned gages lost or damaged without notification	
Out-of-tolerance findings with no impact assessment			

Clause 8.5.1: Control of production and service provision

What your records showed

What was found	Records	Which
Customer requirement not flowed down	2	FLD-MER-03 (Airbus) in Heat Treat, FLD-MER-12 (Airbus) in FAI
FIND-MER-003 Systematic OTD failure — 4 consecutive months below 60% on nacelle bracket program	-	
FIND-MER-011 CNC program revision control failure — obsolete G-code executed on 5-axis center	-	
FIND-MER-006 Rivet spacing nonconformance on fuselage panel lot 26-018 — acceptance criteria not met	-	

	Question	Priority	Evidence location / initials
☐	QB-8.5.1-009 Are special processes (welding, heat treatment, NDT, surface treatment) controlled per qualified procedures and performed by qualified personnel?	Critical	
☐	QB-8.5.1-011 Are FOD (Foreign Object Debris/Damage) prevention controls implemented at all stages of production and assembly?	Critical	
☐	QB-8.5.1-015 Are process validation records maintained for processes whose output cannot be verified by subsequent inspection?	Critical	
☐	QB-8.5.1-019 Are production hold/stop-work procedures defined and is there authority for any employee to stop production for quality?	Critical	
☐	QB-8.5.1-001 Are production controls defined, documented, and applied including work instructions at point of use?	High	

	Question	Priority	Evidence location / initials
☐	QB-8.5.1-008 Are work instructions available and adequate for all operations that affect product quality?	High	
☐	QB-8.5.1-010 Are production processes monitored and controlled using statistical methods where applicable?	High	
☐	QB-8.5.1-012 Are inspection personnel qualified and are inspection methods validated for the types of inspections performed?	High	
☐	QB-8.5.1-013 Are preventive maintenance programs established for production equipment to prevent unplanned downtime and quality issues?	High	
☐	QB-8.5.1-014 Are temporary deviations from approved processes documented, approved, and time-limited?	High	
☐	QB-8.5.1-016 Are controlled production environments (clean rooms, temperature-controlled areas) monitored and alarmed?	High	
☐	QB-8.5.1-017 Are sampling plans used for inspection statistically valid and documented per appropriate standards (e.g., AS13100, ANSI/ASQ Z1.4)?	High	
☐	QB-8.5.1-018 Are automated inspection systems (CMM, vision systems, CT scanning) validated and maintained per requirements?	High	
☐	QB-8.5.1-020 Are temporary repairs on tooling and equipment assessed for impact on product quality?	High	

	Have ready	
☐	Special process qualification records	☐ Process capability (Cpk) trend data
☐	Process parameter monitoring records	☐ Inspector qualification procedure and records
☐	Personnel qualification/certification records	☐ Visual acuity test records
☐	Re-qualification schedule and records	☐ Inspection method validation records
☐	FOD prevention program documentation	☐ Stamp/signature authority register
☐	FOD zone designation records	☐ Preventive maintenance program documentation
☐	Tool accountability records	☐ PM schedule and completion records
☐	Hardware accountability (nut/bolt count) records	☐ Equipment criticality classification
☐	FOD inspection records at assembly close-out	☐ Correlation data between PM and quality performance
☐	FOD incident reporting and investigation records	☐ Temporary deviation/concession procedure
☐	Process validation procedure	☐ Temporary deviation records with approvals and expiry dates
☐	Process validation records	☐ Product traceability for items made under deviations
☐	Re-validation criteria and schedule	☐ Permanent resolution action plans
☐	Process validation review and approval records	☐ Environmental monitoring system records
☐	Stop-work/production hold procedure	☐ Alarm threshold settings documentation
☐	Stop-work authority documentation	☐ Excursion response procedure
☐	Stop-work event records and outcomes	☐ Excursion investigation records
☐	Training records on stop-work authority	☐ Sampling plan documentation referencing standards
☐	Work instructions at workstations	☐ AQL selection justification
☐	Process control documents	☐ Customer-approved sampling plans

	Have ready		
☐	Equipment setup records	☐	Switching rules application records (normal/tightened/reduced)
☐	Traveler/router documents	☐	Automated inspection system validation records
☐	Work instructions at point of use	☐	System calibration and maintenance records
☐	Revision control on work instructions	☐	Correlation study results (automated vs. manual)
☐	Visual aids for complex operations	☐	Program change control records
☐	Operator acknowledgment of instruction review	☐	Temporary repair assessment and approval records
☐	SPC charts for critical characteristics	☐	Temporary repair expiration tracking
☐	Control chart limit calculation records	☐	Product quality monitoring during temporary repair period
☐	Reaction plan documentation for out-of-control conditions	☐	Permanent repair action plan

Red flags an auditor looks for	
Special process performed without qualification	Inspectors without current qualifications performing inspections
Unqualified personnel performing special processes	No visual acuity testing program
No FOD prevention program in assembly areas	No preventive maintenance program exists
FOD found in delivered products	Critical equipment PM consistently overdue
No tool accountability for open-cavity assembly	Temporary deviations expired but still in use
Critical processes not validated	No formal approval process for temporary deviations
Validation records missing or incomplete	Environmental monitoring gaps during non-working hours
No stop-work authority defined for production personnel	Excursions with product in area and no impact assessment
Operators express fear of consequences for stopping production	No documented basis for sampling plans
No work instructions for critical operations	100% inspection required by customer but sampling used instead
Obsolete instructions being used	CMM programs unvalidated or significantly different from manual results
Operators working from outdated instruction revisions	No program change control for automated inspection
SPC charts consistently show out-of-control with no action	Long-standing temporary repairs with no permanent fix plan
Control limits based on specification, not process data	Quality issues linked to equipment in temporary repair status

Clause 8.5.2: Identification and traceability

What your records showed

What was found	Records	Which
Customer requirement not flowed down	1	FLD-MER-02 (Boeing) in Receiving Inspection
FIND-MER-001 Ti-6Al-4V traceability failure — 3 forging lots with incorrect heat lot on CoC	-	

	Question	Priority	Evidence location / initials
☐	QB-8.5.2-002 Is product identification maintained throughout all stages of production including lot/batch/serial traceability?	Critical	
☐	QB-8.5.2-004 Is material traceability maintained from raw material receipt through finished product delivery?	Critical	
☐	QB-8.5.2-001 Is product identification and traceability maintained throughout production?	High	
☐	QB-8.5.2-003 Is the inspection and test status of product clearly indicated at all stages of production?	High	

Have ready	
☐ Product identification procedure	☐ Traceability records (lot/batch tracking)
☐ Traceability records linking product to materials and processes	☐ Material certifications linked to product
☐ Marking/labeling specifications	☐ Inspection status marking/identification
☐ Lot/batch/serial number assignment records	☐ Traveler/router with traceability data
☐ Traceability procedure	☐ Inspection status identification procedure
☐ Material traceability records (heat/lot through production)	☐ Status indicators in use (stamps, tags, color codes, locations)
☐ Reverse traceability demonstration records	☐ Work-in-progress staging area controls
☐ Sub-tier traceability requirements	☐ Inspection hold point enforcement records

Red flags an auditor looks for	
Product found without identification in production area	No traceability system for critical materials
Traceability breaks - cannot link product to material certs	Unable to trace product to material source
Cannot trace finished product back to raw material	Product shipped without clear evidence of inspection completion
Traceability breaks at key manufacturing stages	Mandatory hold points bypassed

Clause 7.2: Competence

What your records showed

What was found	Records	Which
FIND-MER-005 12 expired operator certifications on active production lines	-	

	Question	Priority	Evidence location / initials
☐	QB-7.2-002 Are competency requirements defined for personnel performing special processes (welding, NDT, heat treatment)?	Critical	
☐	QB-7.2-006 Is initial and recurrent training performed for personnel involved in handling hazardous materials, dangerous goods, or ITAR-controlled items?	Critical	
☐	QB-7.2-001 Has required competence for personnel affecting quality been determined?	High	
☐	QB-7.2-004 Are competency matrices maintained showing current qualification status for all personnel in quality-affecting roles?	High	
☐	QB-7.2-005 Are personnel qualified through a combination of education, training, and experience as required by the role?	High	
☐	QB-7.2-007 Are temporary or agency workers given adequate quality training and supervision before performing quality-affecting work?	High	
☐	QB-7.2-003 Is training effectiveness evaluated beyond simple attendance tracking?	Medium	

Have ready	
☐ NDT personnel certification records (Level I, II, III)	☐ Action plans for competency gaps
☐ Welder qualification records per applicable code	☐ Qualification criteria for key roles
☐ Special process operator qualification records	☐ OJT program documentation and completion records
☐ Proficiency testing or re-examination records	☐ Mentoring program records
☐ Dangerous goods training records and certifications	☐ Experience verification records for new hires
☐ ITAR awareness training records	☐ Temporary worker orientation and training records
☐ Hazardous material handling training records	☐ Supervision requirements for temporary workers
☐ Recurrent training schedule and compliance tracking	☐ Competence verification records for temporary workers
☐ Competence matrices or skills matrices	☐ Restricted operations list for temporary workers
☐ Training records	☐ Training effectiveness evaluation criteria
☐ Qualification records for special process operators	☐ Post-training assessment records
☐ Auditor qualification records	☐ On-the-job training validation records
☐ Competency/skills matrix for production and quality roles	☐ Training gap analysis documentation
☐ Qualification status tracking (current, expiring, expired)	

Red flags an auditor looks for	
NDT personnel performing inspections without current certification	Personnel performing tasks for which they are not qualified
No qualification records for special process operators	No OJT program for production roles
Eye exam requirements not met for visual inspection personnel	Personnel performing critical operations without verified competence
Dangerous goods shipped by untrained personnel	Temporary workers performing unsupervised critical operations
ITAR-controlled items handled by persons without training	No quality training provided to temporary workers
Personnel performing critical tasks without documented competence	No training effectiveness evaluation exists
No qualification records for NDT or special process operators	Operators performing tasks for which training effectiveness was not confirmed
No competency matrix exists	

Clause 8.1.1: Operational risk management (AS9100 specific)

What your records showed

What was found	Records	Which
High-priority failure mode with no control	6	MER-3104 (PFMEA-MER-3104), MER-3103 (PFMEA-MER-3103), MER-3101 (PFMEA-MER-3101), MER-3105 (PFMEA-MER-3105), MER-3106 (PFMEA-MER-3106), MER-3102 (PFMEA-MER-3102)

	Question	Priority	Evidence location / initials
☐	QB-8.1.1-001 Is operational risk management applied throughout product realization?	Critical	
☐	QB-8.1.1-002 Is operational risk management integrated into production planning, including critical items and key characteristics?	Critical	
☐	QB-8.1.1-004 Are control plans established for production of critical and safety-critical parts?	Critical	
☐	QB-8.1.1-003 Are project management disciplines applied to product realization including schedule, resource, and risk management?	High	
☐	QB-8.1.1-005 Are advanced product quality planning (APQP) or equivalent processes used for new product introduction?	High	
☐	QB-8.1.1-006 Are manufacturing readiness levels (MRL) or equivalent maturity assessments used for new or transferred production?	Medium	

Have ready	
☐ Operational risk assessments	☐ Project plans with milestones and deliverables
☐ Product safety risk analysis	☐ Resource allocation plans
☐ Risk mitigation records throughout product lifecycle	☐ Project risk registers
☐ Management of operational risk changes	☐ Schedule tracking and variance reports
☐ Critical items list with traceability to design	☐ APQP or equivalent NPI procedure
☐ Key characteristics identification records	☐ Phase gate review records

Have ready			
☐	Control plans for critical items	☐	PPAP or equivalent production approval records
☐	Work instructions referencing critical/key characteristics	☐	Cross-functional team meeting records
☐	Control plans for critical parts	☐	Manufacturing readiness assessment records
☐	Control plan linkage to PFMEA	☐	Production readiness review checklists
☐	Sampling plans referenced in control plans	☐	Low-rate production results and lessons learned
☐	Reaction plans for out-of-control conditions	☐	Transition plan from prototype to full production
Red flags an auditor looks for			
No operational risk management process		No project management discipline applied to product realization	
Product safety not addressed in risk assessments		Projects consistently late with no root cause analysis	
Operators unaware of critical items they are manufacturing		No structured NPI process for complex aerospace products	
No critical items list despite safety-critical products		New products launched without production approval	
No control plans for safety-critical parts		Full-rate production started without readiness assessment	
Control plan not updated after significant process changes		Significant quality issues at full-rate that were not seen at low-rate	

Clause 8.4.3: Information for external providers

What your records showed

What was found	Records	Which
Customer requirement not flowed down	4	FLD-MER-13 (Spirit) to Great Bend Machining, FLD-MER-11 (Airbus) to Titan Bar & Billet, FLD-MER-01 (Boeing) to Great Plains Titanium, FLD-MER-15 (GE) to Emporia Anodizing Co
Customer requirement only partly flowed down	1	FLD-MER-16 (Boeing) to Aero Fastener Brokers

	Question	Priority	Evidence location / initials
☐	QB-8.4.3-001 Is verification of externally provided products and services adequate?	High	
☐	QB-8.4.3-002 Is receiving inspection adequate to detect nonconforming externally provided product including verification of certifications?	High	
☐	QB-8.4.3-003 Is source inspection performed at critical suppliers when required by customer or risk assessment?	High	

Have ready			
☐	Incoming inspection procedure	☐	Material certification verification records
☐	Incoming inspection records	☐	Skip-lot inspection criteria and qualification records
☐	Certificates of conformance from suppliers	☐	Source inspection requirements register
☐	Source inspection records (if applicable)	☐	Source inspection reports
☐	Supplier nonconformance reports	☐	Customer source inspection requirements flow-down
☐	Receiving inspection procedure	☐	Source inspector qualification records

	Have ready		
☐	Receiving inspection records with results		
Red flags an auditor looks for			
No incoming inspection performed on safety-critical items		Certifications not checked for completeness or correctness	
Supplier CoC accepted without any verification		Customer source inspection requirements not met	
Material released to production before receiving inspection		Source inspection never performed despite poor supplier performance	

Clause 10.2: Nonconformity and corrective action

What your records showed

What was found	Records	Which
Corrective action past due	4	CAPA-MER-019, CAPA-MER-052, CAPA-MER-029, CAPA-MER-042

	Question	Priority	Evidence location / initials
☐	QB-10.2-002 Are corrective actions proportionate to the significance of the nonconformity and its potential impact on product safety?	Critical	
☐	QB-10.2-008 Are escaped nonconformities (customer returns, field failures, warranty claims) prioritized for corrective action?	Critical	
☐	QB-10.2-001 Is there a documented corrective action process with root cause analysis?	High	
☐	QB-10.2-003 Is root cause analysis methodology effective and are multiple root cause analysis tools used appropriately?	High	
☐	QB-10.2-004 Are corrective actions verified for effectiveness and are nonconformities confirmed as not recurring?	High	
☐	QB-10.2-005 Are customer-initiated corrective actions (SCARs) responded to within contractual timelines with adequate root cause analysis?	High	
☐	QB-10.2-006 Is the corrective action process integrated with the nonconformance process to ensure all significant NCRs trigger corrective action?	High	
☐	QB-10.2-007 Is the 8D problem-solving methodology or equivalent used for complex or customer-reported nonconformities?	High	
☐	QB-10.2-009 Are corrective action timelines monitored and is there an escalation process for overdue corrective actions?	High	

	Have ready		
☐	Corrective action procedure with severity classification	☐	Recurrence monitoring data post-corrective action
☐	Extent-of-condition analysis records	☐	Re-opened CAR records when effectiveness not demonstrated
☐	Containment action records	☐	SCAR tracking log with response timelines
☐	Corrective action verification records proportionate to risk	☐	SCAR response quality metrics
☐	Customer escape tracking and prioritization records	☐	Customer acceptance/rejection of SCAR responses

Have ready			
☐	Immediate containment actions for escapes	☐	Root cause analysis in SCAR responses
☐	Customer notification records for escaped nonconformities	☐	NCR-to-CAR escalation criteria
☐	Escape analysis and detection improvement records	☐	NCR-to-CAR conversion tracking
☐	Corrective action procedure	☐	Repeat NCR analysis and automatic escalation records
☐	Corrective action records (CARs)	☐	Customer escape analysis and corrective action records
☐	Root cause analysis records	☐	8D or equivalent problem-solving reports
☐	Effectiveness verification records	☐	Team formation records for problem solving
☐	Supplier corrective action requests (SCARs)	☐	Interim containment action records (D3)
☐	Root cause analysis procedure	☐	Verification of corrective action (D6) records
☐	RCA tool usage examples (5-Why, Ishikawa, fault tree)	☐	Corrective action timeline requirements
☐	RCA training records	☐	Corrective action aging report
☐	Evidence of systemic root causes identified and addressed	☐	Escalation procedure for overdue CARs
☐	Effectiveness verification procedure with timing criteria	☐	Management review of overdue CARs
☐	Effectiveness verification records with evidence		
Red flags an auditor looks for			
Safety-related nonconformities treated same as minor issues		Same nonconformity keeps recurring despite 'effective' corrective actions	
No containment actions while product is at risk		SCARs consistently late without customer communication	
Customer escapes with no corrective action		High SCAR rejection rate indicating poor root cause analysis	
No analysis of detection effectiveness after escapes		Significant NCRs closed without corrective action	
Recurring nonconformities with ineffective corrective actions		Repeat nonconformances not triggering corrective action	
No root cause analysis performed		No structured problem-solving for complex issues	
Corrective actions overdue by more than 60 days		8D reports completed as formality with no real analysis	
Root cause consistently identified as 'operator error' or 'training'		Large percentage of CARs overdue with no escalation	
No formal RCA methodology used		Overdue CARs including safety-related items	
No effectiveness verification performed			

Clause 8.1: Operational planning and control

What your records showed

What was found	Records	Which
Customer requirement only partly flowed down	3	FLD-MER-18 (Airbus) in Calibration, FLD-MER-10 (GE) in All special processes, FLD-MER-09 (Airbus) in Heat Treat

	Question	Priority	Evidence location / initials
☐	QB-8.1-002 Are operational planning outputs documented and sufficient to demonstrate product conformity at each stage?	High	
☐	QB-8.1-001 Are operational processes planned with defined criteria, resources, and controls?	Medium	

Have ready	
☐ Manufacturing plans / route cards / travellers	☐ Production control plans
☐ Inspection and test plans	☐ Work order documentation
☐ Acceptance criteria for each operation	☐ Process flow diagrams
☐ Product safety planning outputs	☐ Resource planning records

Red flags an auditor looks for	
No defined acceptance criteria for in-process operations	No formal production planning process
Product safety characteristics not identified in planning	

Clause 8.6: Release of products and services

What your records showed

What was found	Records	Which
Customer requirement not flowed down	1	FLD-MER-05 (GE) in Heat Treat + Inspection
FIND-MER-010 Engine mount bracket hardness below spec — product released without proper disposition	-	

	Question	Priority	Evidence location / initials
☐	QB-8.6-002 Are final inspection and test requirements defined and are all planned verification activities completed before product release?	Critical	
☐	QB-8.6-003 Is the Certificate of Conformance (CoC) or equivalent release documentation accurate and complete before shipment?	Critical	
☐	QB-8.6-001 Are product release criteria defined and applied, with authorization by competent personnel?	High	

	Question	Priority	Evidence location / initials
☐	QB-8.6-004 Are customer witness and hold points defined in inspection plans and properly managed?	High	

Have ready	
☐ Final inspection procedure	☐ Release procedure
☐ Final inspection records showing all checks completed	☐ Authorization matrix for release personnel
☐ Test results with pass/fail status	☐ Completed inspection/test records
☐ Release authorization signatures	☐ Certificate of conformance
☐ Certificate of Conformance (CoC) or equivalent	☐ Release documentation (traveler sign-offs)
☐ CoC template and content requirements	☐ Customer witness/hold point register
☐ Sample CoCs verified for accuracy	☐ Customer notification records for witness points
☐ CoC signatory authorization records	☐ Customer attendance or waiver records
☐ Customer-specific CoC requirements register	☐ Hold point enforcement records in travelers

Red flags an auditor looks for	
Product released with outstanding test requirements	Product released without completing all required inspections
No formal release authority - anyone can ship product	No defined release authority
False or inaccurate information on CoCs	Work progressed past hold point without customer involvement
CoCs signed by unauthorized personnel	Customer witness/hold points unknown to production

Clause 7.5: Documented information

What your records showed

What was found	Records	Which
FIND-MER-008 Pyrometry control procedure (PYR-200) obsolete — operators working without valid WI	-	

	Question	Priority	Evidence location / initials
☐	QB-7.5-004 Are customer-supplied documents (drawings, specifications, standards) controlled for current revision?	Critical	
☐	QB-7.5-001 Is documented information controlled per AS9100 requirements?	High	
☐	QB-7.5-002 Are electronic records and signatures controlled to ensure authenticity, integrity, and prevent unauthorized modification?	High	
☐	QB-7.5-003 Are record retention requirements defined and do they meet customer, regulatory, and statutory obligations?	High	
☐	QB-7.5-005 Are documented procedures maintained for all QMS processes and are they accessible to personnel who need them?	Medium	

Have ready			
☐	Customer document revision tracking system	☐	Audit trail samples showing change history
☐	Current revision verification records	☐	Backup and recovery test records
☐	Obsolete customer document withdrawal records	☐	Record retention schedule with regulatory references
☐	Customer notification subscription records	☐	Storage condition monitoring records (if physical records)
☐	Document control procedure	☐	Record destruction authorization records
☐	Master document list / document register	☐	Customer-specific retention requirements register
☐	Evidence of controlled distribution	☐	QMS procedure register or master list
☐	Record retention schedule	☐	Procedure review schedule and records
☐	Backup and recovery procedures for electronic records	☐	Accessibility verification at point of use
☐	Electronic records management procedure	☐	Procedure training records for affected personnel
☐	Electronic signature validation records		
Red flags an auditor looks for			
Production using outdated customer drawing revision		Backup recovery never tested	
No tracking system for customer document revisions		Records destroyed before regulatory retention period ends	
Obsolete documents in active use on shop floor		No defined retention periods for some record types	
No backup system for electronic quality records		Records stored in conditions causing deterioration	
Shared accounts used for quality record entries		Critical processes operating without documented procedures	
No audit trail on electronic quality records		Actual practice significantly different from documented procedure	

Clause 9.2: Internal audit

What your records showed

What was found	Records	Which
FIND-MER-009 Internal audit program lapsed — no internal audits for 4 months (May to September) during QM vacancy	-	

	Question	Priority	Evidence location / initials
☐	QB-9.2-001 Is an internal audit program planned and executed covering all QMS processes?	High	
☐	QB-9.2-002 Is the internal audit program risk-based, covering all QMS processes including aerospace-specific requirements?	High	
☐	QB-9.2-004 Are internal audit findings effectively addressed with root cause analysis and verified corrective actions?	High	
☐	QB-9.2-008 Are special unannounced audits conducted in response to quality events, customer complaints, or product safety issues?	High	
☐	QB-9.2-003 Are internal auditors qualified, independent, and following a systematic audit methodology?	Medium	

	Question	Priority	Evidence location / initials
☐	QB-9.2-005 Do internal auditors use process-based audit techniques rather than purely clause-based approaches?	Medium	
☐	QB-9.2-006 Is internal audit performance itself evaluated (coverage, timeliness, finding quality, value added)?	Medium	
☐	QB-9.2-007 Are supplier quality audits integrated into the internal audit program or managed as a complementary program?	Medium	

	Have ready		
☐	Internal audit program/schedule	☐	Auditor qualification criteria and records
☐	Audit reports	☐	Auditor training records (lead auditor, AS9100 standard)
☐	Auditor qualification records	☐	Audit plan and checklist examples
☐	Corrective action records for audit findings	☐	Independence verification records
☐	Audit closure verification records	☐	Process-based audit checklists
☐	Audit program with risk-based scheduling	☐	Product audit procedure and records
☐	Audit schedule showing coverage of all AS9100D clauses	☐	Audit plans showing process-based approach
☐	Audit reports for aerospace-specific requirements	☐	Turtle diagram-based audit records
☐	Product audit reports	☐	Audit program effectiveness metrics
☐	Audit finding corrective action records	☐	Audit coverage tracking data
☐	Root cause analysis for each finding	☐	Auditor performance feedback records
☐	Corrective action effectiveness verification records	☐	Audit program improvement actions
☐	Audit finding closure timeline tracking	☐	Supplier audit program or schedule
☐	Event-triggered audit procedure	☐	Supplier audit reports
☐	Records of unplanned audits and their triggers	☐	Supplier audit results in supplier performance tracking
☐	Rapid response audit reports	☐	Supplier audit finding closure tracking
☐	Event-triggered audit findings and corrective actions		

Red flags an auditor looks for	
Some processes never audited	No auditor qualification criteria defined
Audit findings open for over 12 months	Auditors auditing their own work areas
Auditors auditing their own work	Audits purely paper-based with no shop floor verification
AS9100D-specific clauses not covered in audit program	No process-based or product-based audit approach used
Audit findings recurring year after year	Audit program never evaluated for effectiveness
Audit findings recurring year after year with same root cause	Audits finding no issues despite known quality problems
No effectiveness verification on corrective actions	No supplier audit program despite critical supply chain
Significant quality events with no targeted audit response	Supplier audit results ignored in supplier decisions
No capability for event-triggered audits	